

LATENTSPRING: PHYSICS-INFORMED MOLECULAR FLOW MATCHING FROM ATOMIC COMPOSITION

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ABSTRACT

Generating a molecule from its atomic composition requires deciding both how its atoms are arranged in space and which atoms are bonded. LatentSpring combines flow matching with a harmonic-mixture source and learned physical correction. We construct the source without a known bond graph: a random tree organizes Gaussian displacements whose scales depend on atomic radii. Averaging over trees gives a normalized distribution of correlated coordinates. A bounded equivariant head then learns from forces evaluated at the generator’s provisional final geometries, while the backbone remains fixed. Expressing the correction as an endpoint change also lets it act on a diffusion model’s noise prediction. We additionally study transferring physical feedback into the backbone weights. On 64 unseen compositions with 17–64 atoms, the harmonic source improves chemical-graph validity by 3.89 percentage points on average across five fine-tuning runs, with gains in four runs. On 16 further compositions, an 8,178-parameter force head increases the fraction of outputs that are both graph-valid and have GFN2-xTB force RMS at most 5 eV/Å from 38.87% to 46.88%. Under equal physical-supervision budgets on a separate EGNN backbone, the correction improves this joint yield by 7.03 points for FM and 6.64 points for GAGA on another 16 new compositions, with both fits improving in each case. The corrected flow uses 128 backbone and 64 small-head evaluations per output, without energy evaluations or geometry optimization during generation.

1 INTRODUCTION

An atomic composition tells us which atoms a molecule contains, but not how they are connected or arranged in three dimensions. We study generation from this input, together with the molecule’s total charge and spin multiplicity. The model returns atomic coordinates. We then infer bonds from those coordinates to evaluate the generated structure. Unlike conformation generation with a known molecular graph, this task leaves both connectivity and geometry to be determined.

Flow matching learns to transform samples from a starting, or *source*, distribution into molecular structures (Lipman et al., 2023). An isotropic Gaussian is a simple choice, but it places all atoms on a common spatial scale without using atom-specific geometric information. Harmonic sources instead correlate atom positions through virtual springs. Existing graph-conditioned methods can construct these springs from known chemical bonds. The question here is how to obtain a useful structured source when those bonds are not given.

LatentSpring uses a random tree to organize the starting coordinates. We sample Gaussian displacements along its edges, with scales determined by the connected atoms’ radii, and obtain atom positions by adding these displacements along the tree. A new tree is drawn for every sample. Its edges are not proposed chemical bonds: the tree is used only for initialization, and the flow can generate a structure with different connectivity, including rings. Averaging over the trees gives a normalized mixture of harmonic networks. We combine this source with an equivariant coordinate network and geometry self-conditioning: the network makes a provisional geometry prediction, then uses it to refine its velocity prediction.

We incorporate physical feedback through a small learned correction. The frozen parent predicts a provisional final geometry, and an energy model supplies forces at that geometry during training. A bounded equivariant head learns the additional velocity needed to shift the prediction along those

forces. At generation time, only the neural networks are evaluated. Expressing the head’s action as a change to the provisional endpoint also allows it to correct a diffusion model’s noise prediction. A complementary implementation transfers physical feedback into the backbone weights by subtracting a matched *replay* update from an update trained on force-shifted structures.

Our contribution is a harmonic-mixture source for composition-conditioned generation, its integration with coordinate flow matching and two forms of physical learning, and controlled experiments that test these choices. We compare sources, remove self-conditioning, and test physical correction on independently trained flow and diffusion backbones. Unseen compositions and two energy models measure the quality of unoptimized outputs.

2 RELATED WORK

Molecular flows and source distributions. ConfGF and GeoDiff generate conformations for a supplied chemical graph (Shi et al., 2021; Xu et al., 2022). We use FlowMol3’s equivariant coordinate network and geometry self-conditioning modules (Dunn & Koes, 2026). Equivariance means that rotating the input rotates the predicted vectors, while relabeling identical atoms relabels their outputs. HarmonicFlow combines harmonic sources with self-conditioned flow matching for ligand docking; FlowSite extends this approach to binding-site design (Stark et al., 2024). ET-Flow also builds a harmonic source from a supplied molecular graph (Hassan et al., 2024). These methods know the chemical connectivity at the input. Our source instead averages over auxiliary spatial trees and requires only atomic composition. Other approaches to source adaptation include guidance with a frozen vector field (Wang et al., 2026b) and learned condition-dependent sources (Kim et al., 2026).

Distributions over trees. Factored tree distributions allow tractable probabilities and learning over trees (Meila & Jaakkola, 2000; Duan & Dunson, 2023). Gaussian mixtures defined through random spanning-tree differences have also been used for imaging inverse problems (Everink, 2026). We use isotropic edge displacements and atomic length scales to define a distribution over centered molecular coordinates. Our source experiments compare this mixture with isotropic Gaussian, shell-tree, and covariance-matched Gaussian alternatives.

Physical feedback and nonequilibrium work. OMol25 provides molecular structures, energies, and forces across electronic conditions (Levine et al., 2025). Energy-weighted flow matching (Dern et al., 2025), importance-sampling-based training (Midgley et al., 2023), and physical-reward fine-tuning of molecular diffusion models (Zhou et al., 2025) provide ways to incorporate physical information into generation. Simulation data also support protein fine-tuning (Lu et al., 2024), and physical proxies can guide residual correction of frozen video flow fields (Wang et al., 2026a). Our head learns molecular force shifts at predicted endpoints. In a separate extension, we correct the sampling weights of local targets using nonequilibrium work identities. Jarzynski’s equality connects work averages to equilibrium free-energy differences, and Hummer–Szabo reconstruction applies it to single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport includes an imposed coordinate flow and its volume correction (Vaikuntanathan & Jarzynski, 2008); targeted perturbation similarly accounts for coordinate maps in free-energy reweighting (Jarzynski, 2002). Learned escorted protocols have been used for multistate free-energy estimation (Holdijk et al., 2026). We study these corrections with normalized local references, controlled molecular rotors, and subsequent neural training.

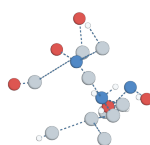
3 A HARMONIC SOURCE WITH LATENT CONNECTIVITY

The input is $c = ((z_i)_{i=1}^N, Q, S)$: the atomic number z_i of every atom, total charge Q in elementary-charge units, and spin multiplicity S . Repeated elements appear once per atom. A neutral singlet has $Q = 0$ and $S = 1$. For example, $\text{C}_2\text{H}_6\text{O}$ specifies two carbon atoms, six hydrogens, and one oxygen, but allows different bond connectivities. Bonds are inferred from coordinates for evaluation, not supplied as training labels.

The output coordinates form $X \in \mathbb{R}^{N \times 3}$, where row x_i is atom i ’s position in angstroms. We remove absolute translation by subtracting the mean position. The coordinates therefore lie in $\mathcal{H}_N = \{X : \sum_i x_i = 0\}$, a space with $3(N - 1)$ independent coordinates.

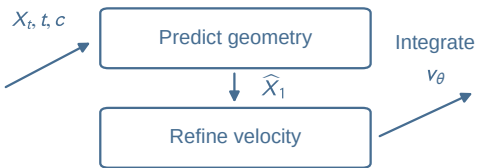
a Generation from atomic composition

Latent spring source



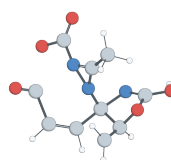
$$X_0 \sim q_0(\cdot | c)$$

Self-conditioned flow



Shared network

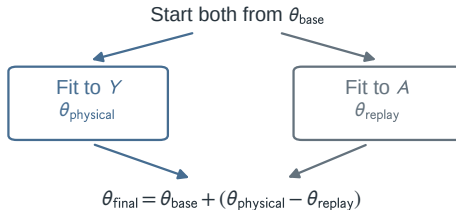
Raw coordinates



$$X_{\text{gen}} \in \mathbb{R}^{27 \times 3}$$

b Force-shifted targets

$$\delta_A = \sigma_A^2 F_+(A) / (k_B T)$$

c Paired fine-tuning

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}})$$

Figure 1: LatentSpring with paired physical training. (a) Source and raw output from a recorded flow for neutral singlet $\text{C}_{10}\text{H}_9\text{N}_3\text{O}_5$. The shared network first forms a provisional geometry, then refines its velocity prediction from current coordinates, time, and composition. Dashed source edges are auxiliary springs; solid output bonds are inferred from coordinates. (b) An actual training anchor and a nearby force-shifted target share a viewing orientation. (c) Two copies learn from the shifted targets and original anchors; their parameter difference is added to the base model. Carbon, hydrogen, nitrogen, and oxygen are gray, white, blue, and red. The coordinates are unoptimized.

Sampling a tree. We first sample a spanning tree T over the atoms: a connected graph with $N - 1$ edges and no cycles. This tree tells us how to place the starting coordinates; it does not specify the final chemical bonds. Each possible edge (i, j) has a positive weight, or affinity, $a_{ij} = a_{ji}$. We sample a tree with probability proportional to the product of its edge weights:

$$p(T | c) = \frac{\prod_{(i,j) \in T} a_{ij}}{\tau(a)}, \quad \tau(a) = \sum_T \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

The sum $\tau(a)$ normalizes this distribution over all spanning trees. We use $a_{ij} = u_i u_j$, where $u_i = 0.05$ for H and F/Cl/Br/I, and $u_i = \max(1, \text{default valence}(z_i) - 1)$ otherwise. Larger u_i favors more tree edges incident to atom i . Small weights for hydrogen and halogens favor terminal positions. Default valence is an element-level bonding capacity, counting bond order; these weights require no molecule-specific bonds.

Placing atoms along the tree. For each tree edge, we independently sample a three-dimensional Gaussian displacement. Its scale depends on ℓ_{ij} , the sum of the two atoms’ covalent radii, which estimates a typical single-bond separation. We set $\sigma_{ij}^2 = \ell_{ij}^2 \exp(s^2)/3$ with $s = 0.2$. The edge displacement $d \in \mathbb{R}^3$ has density

$$h_{ij}(d) = (2\pi\sigma_{ij}^2)^{-3/2} \exp\left[-\frac{\|d\|^2}{2\sigma_{ij}^2}\right]. \quad (2)$$

Here σ_{ij} is the standard deviation along each axis. All directions are equally likely, and the radius-based scale determines the typical separation. This choice also matches the edge covariance of the lognormal shell control. We sample the tree with the weighted Wilson algorithm, add the edge vectors along tree paths to place the atoms, and subtract the mean position. We draw a new tree for every sample. The resulting law is unchanged by rotations or relabelings of identical atoms.

The source density. The tree is unobserved after initialization. Summing over it gives the source density $q_0(X | c)$, a mixture of harmonic networks:

Proposition 1 (Normalized source density). *For normalized isotropic edge densities,*

$$q_0(X | c) = N^{3/2} \frac{\tau(a_{ij} h_{ij}(x_i - x_j))}{\tau(a)} \quad (3)$$

integrates to one over the centered coordinate space $\mathcal{H}_N$.

The numerator sums tree weights after multiplying each edge affinity by its displacement density. The factor $N^{3/2}$ converts independent edge vectors into centered atom coordinates. The matrix-tree theorem evaluates each sum through a determinant rather than enumerating trees; Appendix J derives the normalization. Training only needs samples from the tree-and-edge procedure. It does not evaluate this determinant. The physical teacher introduced below uses a separate local Gaussian reference, q_A .

We call the source *harmonic* because each Gaussian edge density corresponds to a quadratic spring energy. These virtual springs organize the initial random coordinates; the learned flow determines the final molecular geometry.

4 INTEGRATION WITH MOLECULAR FLOW MATCHING

Given c , the flow transforms $X_0 \sim q_0(\cdot | c)$ into $X_{\text{gen}} \in \mathcal{H}_N$. The network takes the current coordinates X_t , flow time t , and c , and predicts $v_\theta(X_t, t; c) \in \mathbb{R}^{N \times 3}$: one velocity vector per atom. Integrating this field from $t = 0$ to $t = 1$ produces the coordinates. The latent tree is used only to sample X_0 and is not passed to the network as a bond graph. Energy and force evaluations are used during physical training, not during generation.

Training pairs and velocity targets. We draw X_0 from the harmonic source and take a reference structure X_1 from OMol25. We align the structures by matching atoms of the same type and rotating them to reduce their coordinate displacement. We then apply the same random rotation and the same random relabeling of identical atoms to both structures. The rotation is uniform over orientations. This shared augmentation preserves the source marginal (Appendix K).

We use the linear path $X_t = (1 - t)X_0 + tX_1$, with velocity target $V^* = X_1 - X_0$. Thus training is supervised regression from an intermediate structure to the displacement between its paired endpoints. The variable $t \in [0, 1]$ measures progress along this interpolation, not physical time.

Geometry self-conditioning. We retain FlowMol3’s coordinate-shaped output head $D_\theta(X_t, t; c)$ and use its increment $D_\theta - X_t$ as the predicted velocity, after subtracting the mean atom velocity to keep coordinates centered. Self-conditioning evaluates the same network twice. The first prediction $v^{(0)}$ gives an estimated endpoint, $\hat{X}_1 = X_t + (1 - t)v^{(0)}$. The second pass receives this provisional geometry as an additional input and predicts $v^{(1)}$. Both passes share weights.

We supervise both velocity predictions:

$$\mathcal{L}_{\text{FM}} = \frac{1}{2} \mathbb{E} \left[\frac{\|v^{(0)} - V^*\|_F^2 + \|v^{(1)} - V^*\|_F^2}{3N} \right]. \quad (4)$$

The expectation averages over reference structures, source samples, and training times. The loss averages the two passes and the $3N$ coordinate errors, giving each molecule equal weight. Gradients pass through both evaluations. Atom types and charge inputs remain fixed; edge inputs always indicate “no bond.” All FlowMol3 source comparisons use this same coordinate-only objective and two-pass architecture.

Sampling. We use 32 midpoint integration steps, with two velocity evaluations per step. In the implemented scheme, the first evaluation uses the current coordinates and the interval’s midpoint time. The second uses the resulting half-step coordinates at that same time. With two self-conditioning passes per velocity evaluation, this requires $32 \times 2 \times 2 = 128$ network forward passes per attempted sample. We then add 0.025 Å centered Gaussian noise. All FlowMol3 source comparisons use this

sampler, with no subsequent energy-based selection or geometry optimization. Reported metrics use these raw outputs.

4.1 PHYSICAL FEEDBACK THROUGH PAIRED TRAINING TARGETS

We next fine-tune the generator using nearby structures constructed from physical forces. A frozen harmonic generator produces training structures, called *anchors* A , on compositions excluded from evaluation. For each graph-valid anchor, we draw nearby coordinates from $q_A = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. The width σ_A controls the size of the perturbations within the centered coordinate space.

The teacher is a target-construction procedure based on the learned eSEN energy model. We average energies and forces over mirror-related coordinates, giving E_+ and F_+ (Appendix N). Each perturbed structure is shifted by the same anchor force: $Y = X + \delta_A$, where $\delta_A = \sigma_A^2 F_+(A)/(k_B T_{\text{phys}})$ and $T_{\text{phys}} = 300$ K. Here k_B is Boltzmann’s constant. This *force escort* is motivated by a local approximation: if energy is linear near A , weighting the Gaussian by a Boltzmann factor shifts its mean by exactly δ_A . We bound the displacement by choosing σ_A before drawing candidates. We retain only candidates that pass the geometric checks and preserve the anchor’s inferred labeled graph, then sample uniformly from them. Appendix N gives the bounds, restriction, and derivation.

We fine-tune two copies of the base weights θ_{base} . The physical copy learns from the force-shifted targets, producing θ_{physical} . The replay copy learns from the original anchors, producing θ_{replay} . Both also train on the same OMol25 references and follow the same schedule. Replay provides a control for further training on generated structures without the physical displacement. We form the final weights as

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}}). \quad (5)$$

This parameter arithmetic (Ilharco et al., 2023) is applied elementwise with coefficient one. It changes the update direction rather than simply retaining the directly fine-tuned physical model. The result is one network, not an ensemble. Section 6.5 and Appendix N compare it with direct fine-tuning and a norm-matched control.

Distributional extensions. We also study *complete-work* weights, which account for the energy change, the local Gaussian density, and any volume change introduced by the coordinate map. Appendices M–O derive these weights and test recovery of specified local distributions. The paired-update generator uses the simpler, uniformly sampled force-shifted targets. Recovering a particular local distribution and reducing generated energy are different objectives.

4.2 LEARNING THE VELOCITY CORRECTION DIRECTLY

The learned-head implementation keeps the backbone fixed and trains a small correction network. At coordinates X_t , the parent prediction gives the provisional endpoint $H_t = X_t + (1-t)v_{\text{base}}(X_t, t; c)$. We query the energy model’s force at H_t during training and use $\delta(H_t)/(1-t)$ as the target additional velocity. A symmetric pair network receives X_t , H_t , atom types, and time, and adds opposite updates to the two atoms of each pair. Generation integrates $v_{\text{base}} + A_\phi$, where A_ϕ is the learned correction. The head starts at zero and preserves the coordinate centroid. Its 20,000-update fit changes only 8,178 parameters. This variant uses 128 backbone and 64 head evaluations per output, with no terminal noise or energy calls. Appendix G gives its targets, bounds, and larger-network controls.

5 EXPERIMENTAL DESIGN

Task and training. Our main experiments adapt a pretrained molecular flow to neutral singlet organic compositions. Every composition contains C and H, with elements drawn from H/B/C/N/O/F/Si/P/S/Cl/Br/I. Training covers 8–40 atoms and evaluation extends to 64; the implementation supports up to 200 atoms. The initial two fine-tuning runs start from the same pretrained checkpoint. Each uses 3,000 distinct OMol25 structures and 3,000 optimizer updates. Within a run, source variants share the training data and initial weights. The network has 5,904,369 parameters. Learning rates are $2 \cdot 10^{-5}$ for the backbone and $3 \cdot 10^{-4}$ for self-conditioning modules. Three additional source runs repeat these two data blocks with new random seeds and report all 1,000-, 3,000-, and 6,000-update checkpoints.

All source variants use the same filtered training subset. In addition to the chemical-graph checks, selection requires a spanning scaffold with bounded edge lengths and atom degrees. This scaffold is a data-selection filter, not an input to the model. Appendix B specifies it.

Evaluation sets. We use four composition sets: 12 for development, 10 for an additional test, 24 for a further test, and 64 for a broader size comparison. Exact atom-count checks confirm that the latter three sets are absent from both processed OMol25 corpora. Their reference coordinates are used only to determine eligibility, not for training. These experiments reuse the same two fine-tuned models. The independent EGNN experiment instead trains from two new random initializations. Selection and source-control details are in Appendix C.

Metrics and uncertainty. The 12- and 10-composition sets use 64 samples per composition, and the 24-composition set uses 32. Geometric validity requires a connected contact graph and no severe atom overlap, with distance thresholds of 1.25 and 0.6 times the summed covalent radii, respectively. Chemical-graph validity additionally requires RDKit to infer a connected, sanitized bond graph with the requested total charge. We divide valid outputs by *all attempts*, including failures and validator exceptions. Graph diversity counts distinct inferred connectivities within each composition. Timings include the full generation procedure. Appendix B gives the detailed checks.

Unless stated otherwise, 95% confidence intervals come from paired bootstrap resampling of generation draws, with the trained models and compositions held fixed. We also resample compositions to describe variation across the evaluation set. Physical scores are computed on saved raw coordinates, using the stated charge and spin and approximate energy models.

6 RESULTS

6.1 EFFECT OF THE LATENT SPRING SOURCE

Replacing the isotropic Gaussian source with the harmonic source improves chemical-graph validity by 5.86 percentage points on the 12 development compositions, 8.83 points on the 10 additional compositions, and 8.20 points on the 24-composition set. These comparisons hold the backbone, initialization, and training data fixed. A Gaussian with approximately matched covariance also performs worse. This supports a benefit beyond the tested covariance approximation, without isolating every property of the mixture. Appendix C reports the source, shell-tree, and covariance controls.

6.2 GENERALIZATION ACROSS MOLECULAR SIZES

We test whether the source advantage persists for larger molecules without further training. The evaluation set contains 64 new compositions, with 16 in each of four size ranges spanning 17–64 atoms. None appears in either processed OMol25 corpus or in the earlier evaluation sets. We select them by fixed metadata ranks after the same reference-validity checks, then generate 32 samples per composition in each fine-tuning run.

Across all 64 compositions, graph validity is 24.85% with the Gaussian source and 31.76% with the harmonic source. The pooled harmonic result is higher in every size range (Appendix D). The overall gain is 6.91 percentage points, with a 95% interval of [4.74,9.03] from resampling compositions within size ranges. The two fine-tuning runs improve by 11.18 and 2.64 points, respectively. Validity decreases as molecules become larger, but the aggregate source advantage remains positive beyond the original 17–28-atom test range.

Three further fine-tuning runs bring the 3,000-update comparison to five runs. The average improvement is 3.89 points, positive in four runs, with a between-run standard deviation of 5.27 points. For the three new runs, the average advantage changes from −1.66 points at 1,000 updates to +1.87 at 3,000 and +3.94 at 6,000. All three favor the harmonic source at 6,000 updates. Appendix D.1 reports every run and training budget.

One composition, distinct connectivities

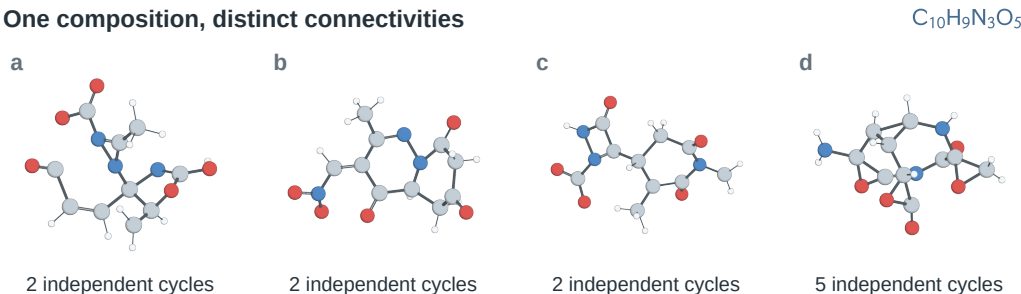


Figure 2: Different connectivities generated from the same atomic composition by the primary force-updated LatentSpring model. The initial trees have no cycles, but the generated molecules can contain rings. We show the first four graph-valid outputs with distinct connectivities for the first CHON composition with valid outputs in run 1 of the 24-composition set. Examples are not selected by energy or altered by geometry optimization. The number of independent cycles is the inferred edge count minus the atom count plus one.

6.3 INDEPENDENT BACKBONE AND GENERATOR OBJECTIVES

We train Gaussian FM, harmonic FM, position-conditional EDM, and GAGA (Hoogeboom et al., 2022; Qu et al., 2026) from scratch on an independent EGNN. All four use the same 20,000 OMol25 structures, architecture, and initial weights within each of two repetitions, with 30,000 updates at batch size 32. This comparison omits self-conditioning and physical adaptation. The two FM models differ only in source distribution; their probability paths and atom pairing are identical. Appendix F gives the protocol and GAGA configuration.

The harmonic source improves FM graph validity by 1.34 percentage points, with a composition-bootstrap 95% interval of [0.12, 2.52]. This supports a source benefit without shared pretraining history. EDM and GAGA achieve higher validity in this setting. Unlike the Gaussian-harmonic FM comparison, those comparisons also change the probability path, training objective, and sampler.

Adding distance self-conditioning to harmonic FM gives 15.63% validity versus 13.77% for validation-selected GAGA on a separate 32-composition panel, at the same training forward-pass budget. The difference is +1.86 points with a composition interval of [-0.34, 4.44]; this comparison does not establish a reliable ordering. Appendix F.1 reports the selection, feedback controls, and physical-update transfer experiment.

The learned physical head transfers to both EGNN generators on another 16 new compositions. At matched physical-supervision budgets, FM joint yield rises from 8.01% to 15.04% and GAGA from 10.94% to 17.58%. Their paired improvements are +7.03 points ([2.73, 11.91]) and +6.64 points ([3.71, 9.77]), respectively, with both fits improving. The ordering between the corrected generators remains uncertain, including after a wider strength calibration (Appendix H).

A further 16-composition comparison evaluates complete systems with their own architectures and training histories. The paired-update model reaches 38.09% GFN2 joint yield, versus 9.77% and 12.11% for GAGA at 128 and 651 calls. Learned force correction gives 46.88%, compared with 38.87% for its frozen parent. Appendix I reports all variants, preparation costs, and the matched-size and training-duration controls. These complete-system results complement the architecture-controlled experiments above.

6.4 NEW COMPOSITIONS AND INDEPENDENT PHYSICAL EVALUATION

We evaluate four frozen models on 24 new compositions: Gaussian, harmonic, harmonic with the force-only update, and harmonic with the complete-work update. The force-only model was selected as the primary physical model before this test. Eight compositions fall in each of the 17–20, 21–24, and 25–28 atom ranges; none appears in either processed corpus. There are 6,144 raw outputs in total. Appendix B describes the fixed metadata-based selection.

Table 1: Graph validity on 24 new compositions, before geometry optimization. Each run has 768 attempts per model, using 128 network forward passes per attempt. The middle columns count valid outputs; the last column pools both runs.

Model	Run 1	Run 2	Pooled rate
Gaussian reference	332	352	44.53%
LatentSpring harmonic	435	375	52.73%
With force update	439	382	53.45%
With complete-work update	426	383	52.67%

Table 2: Effect of the primary force update relative to the frozen harmonic model. The comparison uses paired outputs that are graph-valid under both models. Negative values indicate lower energy or residual force. Both fine-tuning runs improve under both potentials; brackets show paired 95% confidence intervals.

Potential	Energy change (eV/atom)	Force-RMS change (eV/Å)
eSEN	-0.01308 [-0.01805, -0.00817]	-0.34793 [-0.41498, -0.28478]
GFN2	-0.01168 [-0.01616, -0.00724]	-0.35620 [-0.42983, -0.28714]

The harmonic source improves graph validity over Gaussian by 8.20 percentage points (95% CI [5.08, 11.33]), with gains in both runs. Adding the force update changes harmonic validity by +0.72 points, with interval [-0.98, 2.34]. Thus the measured validity rate remains similar after physical fine-tuning. Every accepted output has a distinct inferred connectivity within its composition in these draws.

Energy and force scores. We evaluate every output at its saved coordinates using eSEN and gas-phase GFN2-xTB single-point energies and gradients (Bannwarth et al., 2019). GFN2 is used only for evaluation, independently of the eSEN teacher. Four of the 6,144 generated structures have failed GFN2 calculations; all four are graph-invalid. Calculations succeed for every graph-valid output, and reference inversion checks pass. eSEN uses the mirror averages described in Section 4. Force RMS is the root mean squared magnitude of the atomic force vectors.

The force update lowers energy and force RMS among paired, jointly valid outputs under both potentials; every composition contributes pairs in both runs. The directions also hold across all eSEN outputs and all paired successful GFN2 calculations, regardless of graph validity. Without physical updates, harmonic also lowers aggregate energy relative to Gaussian, predominantly in one run.

Complete-work and force-only updates have similar measured scores. On paired, jointly valid outputs, the complete-work-minus-force energy difference is -0.00413 eV/atom under eSEN (interval [-0.00903, 0.00078]) and -0.00363 under GFN2 ([-0.00792, 0.00075]). Force and graph-validity intervals also include zero. We therefore use the cheaper force-only teacher as the primary method; these comparisons do not resolve an additional benefit from work weighting.

6.5 WHY SUBTRACT THE REPLAY UPDATE

Does replay subtraction help because it changes the update direction, or simply because it reduces the update size? We compare the paired update with direct physical fine-tuning and a direct update rescaled to the same parameter norm. The rescaled control is $\theta_{\text{control}} = \theta_{\text{base}} + \alpha(\theta_{\text{physical}} - \theta_{\text{base}})$, where α is the ratio of paired to direct update norms. Each norm is the square root of the summed squared changes in model parameters; buffers are excluded. The coefficients are 0.2536 and 0.2564 in the two runs, determined from the trained weights before generating evaluation samples.

The rescaled direct update largely preserves graph validity but increases energy relative to the base model. The paired update lowers energy instead. Paired minus same-norm direct gives -0.03232 eV/atom in energy ([-0.03981, -0.02485]) and -0.4170 eV/Å in force RMS ([-0.5529, -0.2859]). Both runs favor the paired direction, and comparisons using all outputs have the same signs. The graph-validity difference is +0.39 percentage points, with interval [-1.95, 2.73]. These results support a benefit from the replay-subtracted direction, rather than update size alone.

Table 3: Update controls on 10 compositions, with 1,280 generation attempts per model. Energy changes are measured with eSEN relative to the base model, using paired outputs that are graph-valid under both models.

Model	Graph validity (%)	Energy change (eV/atom)
Base	62.11	0
Direct physical update	54.77	+0.11080
Same-norm direct update	61.64	+0.01893
Paired update	62.03	−0.01554

Table 4: Source and physical-update comparison under GFN2-xTB, with 1,536 raw attempts per model. Median force RMS uses each model’s own graph-valid outputs with successful calculations. Joint yield uses all attempts and requires both graph validity and force RMS at most 5 eV/Å.

Model	Graph validity (%)	Median force (eV/Å)	Joint yield (%)
Gaussian source	44.53	3.404	35.68
Gaussian + paired update	46.61	3.222	40.63
Harmonic source	52.73	3.287	44.53
Harmonic + paired update	53.45	3.037	47.59

Component controls. At matched training-forward budgets, self-conditioning improves validity by 12.43 points ([9.70, 15.23]). Removing the force displacement raises energy by 0.01939 eV/atom on paired graph-valid outputs. Appendix Q reports both runs, validity, force scores, and the update-matched comparison.

6.6 SOURCE AND PHYSICAL ADAPTATION

We apply paired physical training to both source families, using generated anchors from the same eight training compositions and matched maximum proposal and training budgets. Table 4 compares all four models on 24 compositions. Appendix Q.2 records preparation costs and evaluation reuse.

The paired update improves GFN2 joint yield for both sources: by 4.95 percentage points for Gaussian (composition-bootstrap 95% interval [2.86, 7.10]) and 3.06 points for harmonic ([1.04, 5.14]). Both runs improve for both sources. On paired, jointly valid outputs, the Gaussian update also lowers energy by 0.01465 eV/atom and force RMS by 0.4618 eV/Å.

After physical adaptation, harmonic exceeds Gaussian by 6.84 points in pooled graph validity ([4.23, 9.51]) and 6.97 points in joint yield ([3.26, 10.61]). The validity difference is positive in both runs, but the joint-yield differences are +16.28 and −2.34 points. Thus the pooled advantage coexists with substantial variation between fine-tuning runs. eSEN also shows the within-source benefits of physical adaptation and the pooled harmonic-source advantage. Figure 10 in Appendix Q plots the results across force thresholds.

Learning hydrogen placement. An optional conditional flow refines hydrogen positions while preserving heavy-atom geometry. On 30 new compositions, it raises shared-EGNN FM graph validity from 25.31% to 27.40% and lowers energy by 15.21 meV/atom relative to a fixed radial repair, with similar validity. Moving all hydrogens together improves energy over moving only detached hydrogens. Appendix H.2 reports the model, controls, and equally augmented GAGA comparison.

7 DISCUSSION

LatentSpring combines a latent-tree harmonic source with learned physical correction. The source benefit appears across backbones, and the correction improves both EGNN flow and diffusion generators under equal physical supervision. The results concern unoptimized neutral organic structures and approximate energy models. Work-identity experiments recover specified local distributions; global molecular equilibrium and broader chemistry remain open.

REPRODUCIBILITY STATEMENT

Code and manifests record seeds, checkpoint hashes, evaluation settings, and all attempted outputs. Training, target preparation, and generation costs are reported separately. Tests cover source normalization, symmetry, density gradients, and work identities. Original data and backbone licenses and access requirements apply.

AI USE STATEMENT

OpenAI Codex and Claude Code assisted literature review, hypothesis and method development, experimental design, mathematical analysis and proof drafting, code implementation, data preprocessing, testing, result interpretation, figure construction, and manuscript drafting and editing. Reported numerical results come from executed programs and saved model outputs; the molecular figures use recorded coordinates. The accompanying code tests and result audits document computational checks of the AI-assisted implementation.

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A READER’S GUIDE TO THE NOTATION

LatentSpring first samples starting coordinates from atomic composition, then learns a flow to reference structures, and incorporates physical feedback through paired fine-tuning or an explicit velocity correction. This appendix collects the notation used in those stages.

Inputs and outputs. The condition c fixes the atom identities, charge, and spin. Random starting coordinates X_0 provide variation between generated structures. Each network evaluation takes current coordinates X_t , flow time t , and c and predicts one velocity vector per atom. Self-conditioning uses a first prediction to form a provisional geometry for a second pass. Integrating the velocities and adding the specified terminal noise gives the final coordinate matrix X_{gen} . The learned-correction variant sets this terminal noise to zero. The input atom identities and electronic conditions do not change. Bonds are inferred separately for evaluation.

Structural and physical metrics. Geometric validity requires one connected contact structure with no excessive atom overlap. Chemical-graph validity additionally requires a chemically consistent bond graph inferred from the coordinates. Two paired outputs are *jointly valid* when both pass these checks; they need not have the same inferred graph. This differs from the physical teacher’s *same-graph* restriction, which preserves the anchor’s labeled bond orders, formal charges, and radical counts.

We compare energy within the same atomic composition and report changes per atom. Force RMS is $[N^{-1} \sum_{i=1}^N \|F_i\|_2^2]^{1/2}$, the root mean squared magnitude of the atomic force vectors. Lower force RMS means smaller residual forces. Lower energy means greater energetic preference under the chosen potential. These are different measurements from graph validity.

B DATA SELECTION AND STRUCTURAL EVALUATION

We traverse the processed OMol25 training corpus in a fixed shuffled order and exclude evaluation compositions. The first 3,000 eligible structures form the training set for each fine-tuning run. Eligibility requires the atom-count, element, and electronic-state conditions in Section 5, the structural checks below, and a geometric scaffold filter.

Training-only scaffold filter. For each atom pair, we divide its distance by the sum of the atoms’ covalent radii. Only pairs with normalized distance strictly between 0.6501 and 1.1999 can be scaffold edges. Kruskal’s algorithm builds a minimum normalized-distance spanning tree by adding short edges without creating cycles. We accept a structure only if the tree is connected and its atom degrees are at most 1 for H/F/Cl/Br/I, 4 for B/C/N/Si, 3 for O, and 6 for P/S. This tree filters training coordinates; it is not the random tree used to sample the model’s source.

Symbol	Meaning
N, z_i, Q, S, c	Atom count, atomic numbers, total charge, and spin multiplicity. The condition c collects these inputs.
$X, x_i, \mathcal{H}_N$	Atomic coordinate matrix, the position of atom i , and the space where the mean atomic position is zero. Coordinates are in angstroms.
T, a_{ij}, u_i	Auxiliary spatial tree, its edge affinities, and the atom-dependent factors defining those affinities. These are source variables.
$\ell_{ij}, \sigma_{ij}, h_{ij}$	Characteristic separation from covalent radii, Gaussian displacement scale, and the probability density of one edge vector.
$\tau(a), q_0(X c)$	Sum of tree weights and the normalized distribution of starting coordinates after averaging over trees.
$X_0, X_1, X_t, X_{\text{gen}}, t$	Starting, reference, intermediate and returned coordinates; t labels generation progress, not physical time.
V^*, v_θ, D_θ	Target velocity, predicted velocity, and the coordinate-shaped network head whose centered increment $D_\theta - X_t$ is trained as the velocity.
$v^{(0)}, v^{(1)}, \hat{X}_1$	First and second velocity predictions, and the provisional endpoint passed between them during self-conditioning.
v_{base}, H_t	Frozen parent velocity after both self-conditioning passes, and its extrapolated endpoint used to construct physical training targets.
$A_\phi, \phi, v_{\text{cap}}$	Added velocity, correction-network parameters, and its instantaneous velocity-bound scale.
$\mathbb{E}, \ \cdot\ _F^2$	Average over the specified random draws, and the sum of squared entries of a coordinate or force matrix.
$\theta_{\text{base}}, \theta_{\text{physical}}, \theta_{\text{replay}}, \theta_{\text{final}}$	Base weights, weights after physical or replay fine-tuning, and the final generation weights.
$A, q_A, \sigma_A, \delta_A$	Generated anchor, local Gaussian reference, perturbation width, and force-guided shift.
E_+, F_+	Mirror-symmetrized teacher energy and force, in eV and eV/Å. The force is the negative coordinate gradient of this energy.
$k_B T_{\text{phys}}, W_A$	Thermal energy at the teacher temperature and generalized work used to weight local candidate structures.
Φ, J, Z	Coordinate map, its local volume-change factor, and a target-density normalizing integral.

Table 5: Notation for the source, flow model, and physical update. The tree T and teacher temperature T_{phys} are different quantities.

Inferring and checking bonds. We use RDKit 2025.03.2 for bond inference and chemical-consistency checks. After testing contact connectivity and atom overlap, DetermineBonds assigns bonds at the requested total charge with covFactor=1.25, useVdw=True, allowChargedFragments=True, and embedChiral=False. RDKit sanitization checks valence and related graph rules. An output passes only if sanitization succeeds, formal charges sum to the requested charge, and the inferred graph has one connected component. To measure graph diversity, we remove stereochemical tags and explicit hydrogens and count distinct canonical SMILES, which encode molecular connectivity. Validator errors and rejected structures remain in the generation-attempt denominator.

The 24-composition confirmation set. Before this evaluation, we freeze the Gaussian, harmonic, and two physically updated harmonic models and designate the force-only update as the primary physical model. We start from 256 distinct compositions outside the earlier candidate sets; checking their reference structures leaves 121 eligible compositions. Fixed metadata ranks select eight from each of the 17–20, 21–24, and 25–28 atom ranges. None of the 24 appears in either processed OMol25 training or validation corpus. Reference coordinates are used only to determine eligibility. With 32 samples per composition in each run, the four models produce 6,144 raw outputs.

C SOURCE-DEVELOPMENT AND COVARIANCE COMPARISONS

Composition sets. Checking reference structures leaves 31 eligible compositions out of 78 that pass the metadata conditions. Of these, 21 had appeared in earlier development experiments. We

Table 6: Valid outputs divided by all generation attempts. All variants use the same geometry self-conditioning backbone and 128 network forward passes per output. The additional set evaluates models frozen after the development comparison.

Set	Run	Gaussian	Shell tree	Harmonic tree
Development (12)	1	357/768	417/768	436/768
	2	377/768	342/768	388/768
Additional (10)	1	326/640	—	413/640
	2	349/640	—	375/640

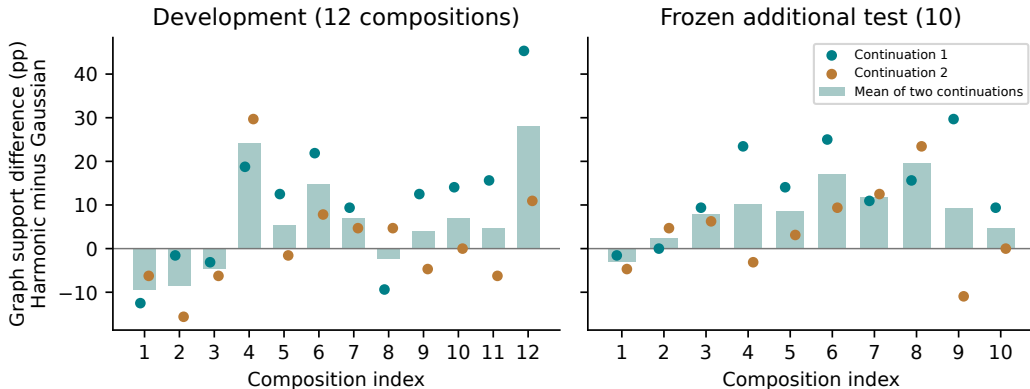


Figure 3: Harmonic-minus-Gaussian graph-validity differences for each composition. Points show the two individual fine-tuning runs, and bars show their mean. Gains vary across compositions and runs. Both evaluation sets use the same trained models.

use a fixed subset of 12 for source development and all 10 remaining compositions for a subsequent test. These 10 have 17–28 atoms and are absent from 3,902,107 processed training rows and 39,415 validation rows, verified by exact atomic-count vectors. Reference geometries only determine eligibility. Both sets use the same two fine-tuning runs; the further 24-composition test is described in Section 6.4.

On the development set, the harmonic source improves graph validity over the isotropic Gaussian by 5.86 percentage points (paired 95% CI [2.67,8.98]; composition-bootstrap interval [0.52,12.17]). Both runs improve, with a larger gain in run 1. Harmonic also exceeds the narrow shell-tree source by 4.23 points; the shell-minus-harmonic interval is [-7.29,-1.11].

On the 10-composition test, harmonic improves graph validity from 52.73% to 61.56%, a gain of 8.83 points (paired 95% CI [5.08,12.50]; composition-bootstrap interval [4.84,12.66]). Geometric validity improves by 9.69 points [6.33,12.89]. Every graph-valid output has distinct inferred connectivity within its composition, and no validator exceptions occur. Generation time is nearly unchanged: harmonic/Gaussian take 175.73/175.43 seconds for 640 samples in run 1 and 178.20/177.74 seconds in run 2.

Control for source covariance. To test whether the source benefit is explained by its covariance, we compare against a single Gaussian with approximately the same covariance as the harmonic mixture. We estimate that covariance from 128 trees and average over permutations of identical atom species. An independent 8,192-tree estimate gives a relative Frobenius error of 1.85–6.25% across the 22 compositions. We match the training data, optimizer, network, and sampling budgets. This follow-up compares the new Gaussian models with the already frozen harmonic checkpoints.

Harmonic improves graph validity over this Gaussian by 11.65 percentage points on development (paired 95% interval [8.59,14.78]; descriptive composition interval [8.07,15.43]) and 15.39 points on the additional set (paired [11.64,19.06]; composition [11.95,18.98]). The difference is positive in both runs on both sets. The covariance-matched Gaussian also performs worse than the isotropic Gaussian.

Table 7: Covariance-matched Gaussian versus harmonic source. Entries count valid outputs over all attempts. Training and inference budgets are matched; the Gaussian uses the calibrated covariance estimate from 128 trees.

Set	Run	Covariance Gaussian	Harmonic tree
Development	1	332/768	436/768
	2	313/768	388/768
Additional	1	286/640	413/640
	2	305/640	375/640

Table 8: Graph validity on the 64-composition set, measured before geometry optimization. Each size range contains 1,024 attempts per method across the two fine-tuning runs. Rates are percentages; differences are percentage points.

Atoms	Gaussian	Harmonic	Difference
17–28	49.12	55.08	+5.96
29–40	24.02	38.28	+14.26
41–52	16.50	21.29	+4.79
53–64	9.77	12.40	+2.64
All 64 compositions	24.85	31.76	+6.91

These results support a benefit beyond this calibrated second-moment approximation, consistent with a contribution from the mixture structure. The comparison adds 2,816 outputs while holding parameter count, initial self-conditioning weights, training structures, and optimization fixed.

D VALIDITY BY MOLECULAR SIZE

The 64-composition evaluation set contains 16 compositions in each size range. The table separates the pooled source comparison by atom count.

D.1 REPEATED SOURCE TRAINING AND BUDGET DEPENDENCE

We add three paired fine-tuning runs from the same pretrained FlowMol checkpoint. Runs 3 and 5 use the first 3,000-row training block; run 4 uses the second. Each pair shares initial weights, reference order, and optimizer settings, and changes only the source. At 6,000 updates, the same 3,000 rows have been visited twice. These are five stochastic fine-tuning runs on two fixed data blocks, not five independent pretraining runs or an isolated dataset-size experiment.

Every new run is evaluated at 1,000, 3,000, and 6,000 updates on the existing 64-composition panel. The first and last checkpoints use 16 draws per composition; the middle checkpoint uses 32, matching the original runs. All checkpoints and all runs are reported. No checkpoint was selected using these outcomes.

Table 9: Harmonic-minus-Gaussian graph validity in percentage points. Dashes mark training budgets not run for the original two models.

Fine-tuning run	1,000 updates	3,000 updates	6,000 updates
1	—	+11.18	—
2	—	+2.64	—
3	−1.37	+5.66	+5.96
4	−0.10	+3.32	+3.91
5	−3.52	−3.37	+1.95
Mean, new runs 3–5	−1.66	+1.87	+3.94
Mean, all available runs	−1.66	+3.89	+3.94

Table 10: Graph validity of the separately adapted conditional EDM, before geometry optimization. Each run has 768 attempts. For context, the harmonic and force-updated harmonic models in Table 1 reach 52.73% and 53.45% at 128 calls, but use a different pretrained backbone.

Sampler	Run 1	Run 2	Pooled rate
EDM, 128 calls	103	100	13.22%
EDM, 1,001 calls	93	106	12.96%

At 3,000 updates, the five-run mean is +3.89 points and the sample standard deviation across runs is 5.27 points. The composition-bootstrap interval is $[2.52, 5.33]$, conditional on these fitted models; it does not describe uncertainty over future training runs. For the three new runs, the corresponding intervals are $[-4.04, 0.68]$, $[0.36, 3.52]$, and $[1.73, 6.25]$ at the three budgets. The source advantage depends on training duration and varies across runs. This study adds 36,000 optimizer updates and 24,576 raw outputs, with no physical-oracle queries.

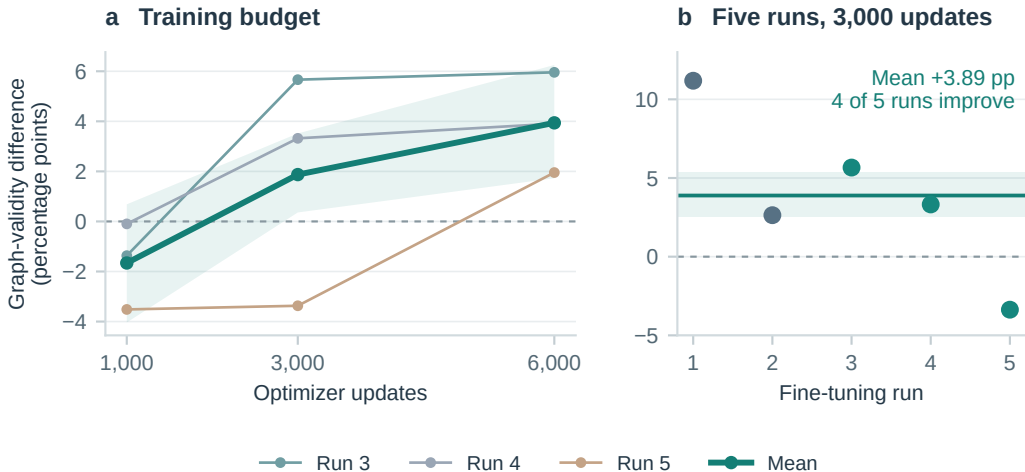


Figure 4: Source effects across training budgets and fine-tuning runs. Thin lines show all three new runs; the thick line is their mean. The right panel includes all five runs at 3,000 updates. Shading and the horizontal bar show 95% composition-bootstrap intervals conditional on the fitted models.

E ADDITIONAL GENERATOR EVALUATIONS

E.1 AN INDEPENDENTLY ADAPTED DIFFUSION BASELINE

We also adapt the official EGNN/EDM backbone (Hooeboom et al., 2022) to composition-conditioned coordinate generation. Atom categories are fixed conditions during training and sampling, while positions follow EDM’s variance-preserving diffusion schedule. Starting from its GEOM-pretrained checkpoint, we train for 30,000 updates on the same 3,000 OMol25 structures used in each source-model run, retaining the original learning rate of 10^{-4} . We evaluate all 24 new compositions with both a 128-call sampler and the full 1,001-call sampler, including the final observation step.

This baseline uses the same task and adaptation data but a different pretrained architecture and objective. EDM has 2.38 million parameters and one denoiser pass per update; LatentSpring has 5.90 million parameters and two passes. The longer EDM fine-tuning runs take 481.51 and 463.61 seconds, compared with about 570–590 seconds for the source-model runs. For 768 outputs, EDM’s 128-call sampler takes 48.54 and 47.94 seconds; its 1,001-call sampler takes 377.65 and 374.12 seconds. Matching network-call counts is therefore not the same as matching computational cost. Pretraining histories also differ, and the composition-overlap checks cover only the processed OMol25 corpora.

Table 11: Graph validity on the 64-composition set for models trained from scratch on the same 2.38-million-parameter EGNN. Each seed has 2,048 generation attempts at 128 network forward passes per output. Counts are graph-valid outputs; the final column pools both seeds.

Method	Valid, seed 1	Valid, seed 2	Pooled rate (%)
Gaussian FM	182	218	9.77
Harmonic FM	223	232	11.11
EDM	301	216	12.62
GAGA	321	249	13.92

We use the matched Gaussian–harmonic experiments, rather than this baseline, to isolate the effect of source choice.

F INDEPENDENT-BACKBONE COMPARISON

This experiment tests the source without pretraining on a second architecture. All methods use the official EDM equivariant graph neural network (EGNN) (Hoogeboom et al., 2022), with four message-passing layers, 256 hidden features, and 2,381,566 parameters. We repeat training from two independent initializations. Within each repetition, all methods start from identical weights and receive the same minibatches. Atom types are fixed inputs throughout training and generation.

Data. The shared data contain 20,000 OMol25 training structures from 8,323 compositions and 512 internal validation structures from 299 other compositions. Both sets contain neutral organic singlets with 8–64 atoms. A hash of composition assigns structures to training or internal validation. Selection requires the geometric and chemical-graph checks, but not the additional scaffold-tree filter used in the pretrained-model experiments. None of the 64 evaluation compositions occurs in either selected subset or either full processed corpus.

Optimization. Each model receives 30,000 AdamW updates at learning rate 10^{-4} and batch size 32, for 960,000 training presentations. Each batch contains molecules with the same atom count. We sample atom-count groups in proportion to their sizes, so each training row has the same marginal sampling probability. We clip the gradient norm at one. Generation uses an exponential moving average of weights at the final update, with decay $\min(0.999, (1 + s)/(10 + s))$ at optimizer step s . Internal validation at 5,000, 15,000, and 30,000 updates monitors learning; it is not used to choose a checkpoint.

Flow-matching models. Gaussian and harmonic FM share the linear interpolation, atom matching, and rotation alignment. Both predict velocity in one EGNN pass and use 64 midpoint steps, giving 128 network evaluations. Neither uses self-conditioning. The harmonic distribution is the same as in the main model, but its factorized edge weights permit a different exact tree sampler. We draw the $N - 2$ entries of a Prüfer sequence independently with probabilities $u_i / \sum_j u_j$ and decode the sequence into a tree. This samples the same tree law used to place atoms.

Diffusion models. EDM predicts Gaussian coordinate noise under its original variance-preserving schedule. Position-conditional GAGA (Qu et al., 2026) truncates this schedule at 650 of 1,000 steps, using the published GEOM setting. Its initial Gaussian has variance $\alpha_t^2 v + \sigma_t^2$, where α_t and σ_t are the signal and noise coefficients. The training-only estimate $v = 3.0232257866 \text{ \AA}^2$ is the mean squared centered coordinate per independent coordinate. We transfer the truncation setting from GEOM without optimizing it on OMol25 evaluation results.

Generation and compute. Every model generates 32 raw structures per evaluation composition at 128 network evaluations. We also evaluate the complete EDM and GAGA reverse schedules at 1,001 and 651 calls, including their final observation steps. This experiment uses no physical energy labels, bond supervision, energy-based selection, or geometry optimization. Scheduler metadata record the A100, H200, and A100 MIG allocations. Equal training presentations and network-call counts do not imply equal wall-clock time.

Table 12: Diffusion results at the shared 128-call budget and the full reverse schedules. Each independent initialization generates 2,048 outputs. Counts give graph-valid outputs; the final rate pools both initializations.

Method	Evaluations	Valid, seed 1	Valid, seed 2	Rate (%)
EDM	128	301	216	12.62
EDM	1,001	308	206	12.55
GAGA	128	321	249	13.92
GAGA	651	273	267	13.18

F.1 GEOMETRY FEEDBACK AND PHYSICAL TRANSFER ON THE EGNN

We test geometry feedback on the independent EGNN using the same 20,000 training rows and two initializations. The first pass predicts provisional coordinates; the second uses their pairwise distances in an existing invariant edge channel. The provisional geometry is detached, and both passes are supervised. Thus this experiment uses 15,000 updates at batch size 32 and two forward passes per example, matching the 960,000 training forward passes of the original one-pass models. The total parameter count is 2,381,566, including 1,001 frozen noise-schedule entries. Optimizer-update counts and wall time are not matched.

An alternative replaces distances with edge-inclusion probabilities from a regularized auxiliary tree distribution. On 32 internal validation compositions, we select distance or tree feedback by pooled graph validity, with one choice for both seeds. GAGA separately selects its maximum sampling timestep from 350, 500, and 650. Validation selects distance feedback and timestep 650. We then evaluate 32 previously ungenerated compositions, eight in each of the four size ranges, with 32 draws per composition and seed. Each output uses 128 network calls, without physical refinement.

Table 13: Raw graph validity on the feedback-confirmation panel. Each model has 1,024 attempts per initialization. All tested feedback choices are shown.

Model	Valid, seed 1	Valid, seed 2	Pooled (%)
Harmonic FM, one pass	107	113	10.74
Harmonic FM, distance feedback	155	165	15.63
Harmonic FM, tree feedback	114	84	9.67
GAGA, selected timestep 650	151	131	13.77

Distance feedback exceeds GAGA by 1.86 points, with a composition-bootstrap 95% interval of $[-0.34, 4.44]$. The seed-specific differences are $+0.39$ and $+3.32$ points. Tree feedback reduces validity relative to distance feedback by 5.96 points ($[-8.11, -3.96]$), so it is not part of the proposed method.

Transferring the fixed physical-update recipe. Each selected parent generates 512 candidate anchors on eight training compositions. We apply the same local force-target procedure, then fit physical and replay students. Each FM student uses 1,000 two-pass updates; each GAGA student uses 2,000 one-pass updates. The learning rate is 10^{-4} and the parameter-difference coefficient is one. This is a follow-up on the same 32-composition panel. It tests this fixed recipe, without selecting its strength on validation.

Table 14: Fixed-strength physical adaptation on the EGNN. Joint yield requires graph validity and GFN2 force RMS at most 5 eV/Å, with all 2,048 attempts per model in the denominator.

Model	Graph validity (%)	Joint yield (%)
FM with distance feedback	15.63	4.79
FM + paired physical update	2.00	0.78
GAGA	13.77	5.86
GAGA + paired physical update	4.79	1.90

The full-strength update lowers both graph validity and joint quality for both families. The within-FM joint-yield change is -4.00 points ($[-6.49, -2.00]$); the within-GAGA change is -3.96 points ($[-6.25, -1.95]$). eSEN gives the same directions. These results show that the successful primary-backbone recipe cannot be transferred unchanged to this EGNN. GAGA’s predefined noise schedule stays fixed in this comparison. Appendix H evaluates a separately trained correction head with broader supervision from each parent’s own trajectory states.

G LEARNING A PHYSICAL CORRECTION ALONG THE FLOW

The paired update incorporates physical feedback into the backbone weights. We also test a separate network that learns the physical correction itself. This separates its regression target from the backbone’s flow-matching error and keeps every parent parameter fixed.

States and targets. The parent generates two trajectories for each of 128 training compositions, with 32 compositions in each of the 8–16, 17–24, 25–32, and 33–40 atom ranges. We record the second velocity evaluation at midpoint steps 23 and 29, corresponding to flow times 0.734375 and 0.921875. These 256 trajectories yield 512 states. At a recorded state, the parent velocity defines the provisional endpoint $H_t = X_t + (1 - t)v_{\text{base}}(X_t, t; c)$. This is an extrapolation from the current prediction after both parent self-conditioning passes; later changes in velocity can change the final generated coordinates.

We evaluate the mirror-averaged eSEN force at H_t , subtract its mean across atoms, and construct the force shift described in Section 4. The Gaussian width is at most $0.03 \text{ \AA}$ and the molecular displacement norm is at most $0.1 \text{ \AA}$. The regression target is $\delta(H_t)/(1 - t)$, a velocity correction whose linear endpoint extrapolation has displacement $\delta(H_t)$. All recorded states are retained, including states whose provisional geometry fails the chemical-graph check. Compositions are separated by metadata hashes: 96 train the head, and 32 provide internal validation, with four recorded states per composition. These are subsets of the parent’s training compositions, separate from generator confirmation conditions.

Equivariant pair correction. The small model has 8,178 parameters. For each unordered atom pair, it computes symmetric scalar features from atomic embeddings, the two pair distances in X_t and H_t , their directional alignment, time, atom count, and covalent radii. A two-hidden-layer network, each layer containing 64 units with SiLU activations, predicts two coefficients α_{ij} and β_{ij} . A hyperbolic tangent bounds both coefficients between -1 and 1 .

Write $r_{ij} = X_{t,i} - X_{t,j}$ and $\hat{r}_{ij} = H_{t,i} - H_{t,j}$. We normalize these vectors as $\bar{r}_{ij} = r_{ij} / \sqrt{\|r_{ij}\|^2 + \ell_{ij}^2}$ and likewise for $\bar{\hat{r}}_{ij}$, using the covalent-radius sum ℓ_{ij} . The contact weight $w_{ij} = \text{sigmoid}((1.25 - \|\hat{r}_{ij}\|/\ell_{ij})/0.2)$ decreases as the predicted atoms move apart. It is a geometric weight, not a supplied bond label. Let $d_i = \sum_{j \neq i} w_{ij}$ and $d_{\max} = \max(1, \max_i d_i)$. The correction for atom i is

$$A_{\phi,i} = v_{\text{cap}} t^2 \sum_{j \neq i} \frac{w_{ij}}{2d_{\max}} \left(\alpha_{ij} \bar{r}_{ij} + \beta_{ij} \bar{\hat{r}}_{ij} \right), \quad v_{\text{cap}} = 2 \text{ \AA per unit flow time.} \quad (6)$$

The symmetric coefficients and opposite pair vectors give zero total correction. Rotating the coordinates rotates the correction, and relabeling atoms relabels its outputs. These pair constructions follow standard equivariant-network practice (Satorras et al., 2021). The triangle inequality gives $\|A_{\phi,i}\| \leq v_{\text{cap}} t^2$ for every atom. This bounds the instantaneous velocity addition, not the final displacement after interaction with the parent flow. The final layer starts at zero, reproducing the parent field before training.

We fit the correction with mean squared velocity error, AdamW at learning rate 3×10^{-4} , batch size one, and gradient clipping at one. Two seeds use 20,000 updates each. Only the head is optimized. We compare a 37,540-parameter pair network with wider layers and a 37,586-parameter alternative with two rounds of neighborhood message passing and invariant angular summaries. All receive the same training-state sequence and target vectors within a seed.

On the internally withheld compositions, the contextual model has about 15% lower target-prediction error than the small pair model and about 11% lower error than the nearly parameter-matched wide

model at 20,000 updates. The confirmation experiment below tests whether this predictive difference improves actual generation. Physics-based residual correction of frozen flow fields has also been studied for video generation (Wang et al., 2026a); here supervision is provided by molecular forces at predicted atomic geometries.

Sampling and preparation cost. Generation integrates $v_{\text{base}} + A_\phi$ with the same 32 midpoint steps. Each attempted structure uses 128 backbone evaluations and 64 head evaluations, with no terminal noise, energy call, output filtering, or geometry optimization. The parent control carries a zero-output head so its sampling operations match. The training-state collection uses 256 parent trajectories. Mirror-paired force labels for the 512 recorded endpoints require 1,024 eSEN queries. The study also queries eight translated Gaussian particles per state for a complete-work control, bringing the actual preparation expenditure to 9,216 queries. The force-target heads do not use those particle energies. The local-work teacher has median particle effective sample size 1.37 out of eight; its earlier 2,000-update student comparison did not establish an additional generation gain.

H PHYSICAL CORRECTION ON A SHARED BACKBONE

We test whether the learned physical correction transfers to the independent EGNN, and give GAGA the same correction architecture and physical-supervision budget. Both parents have 2,381,566 parameters and use the same 20,000 OMol25 training rows. The FM parent uses distance self-conditioning. Its 15,000 two-pass updates match the 960,000 training-example forward passes of GAGA’s 30,000 one-pass updates, as described in Appendix F.1. Two independent parent initializations each receive a separately fitted head.

A common predicted endpoint change. FM predicts a velocity, whereas GAGA predicts noise. We express the head’s action through the change it induces in the parent’s provisional final geometry. Write p for the fraction of the sampling schedule already completed. For FM, p is the flow time; for GAGA, $p = 1 - t/t_{\text{max}}$, where t is diffusion noise time and $t_{\text{max}} = 0.65$ is the selected truncation. These are normalized schedule positions, not a shared physical time.

The head A_ϕ takes the current coordinates X , provisional final geometry H , atom identities, and progress p . It predicts the same bounded vector correction as in Appendix G. A strength η scales its implied endpoint displacement to $\eta(1 - p)A_\phi$. The corresponding updates to the two parent outputs are

$$H = X + (1 - p)v, \quad v^{\text{corr}} = v + \eta A_\phi, \quad (7)$$

$$H = \frac{X - \sigma_t \epsilon}{\alpha_t}, \quad \epsilon^{\text{corr}} = \epsilon - \eta \frac{\alpha_t(1 - p)}{\sigma_t} A_\phi. \quad (8)$$

Here v is the FM velocity, ϵ is GAGA’s predicted noise, and α_t, σ_t are the signal and noise scales of its fixed schedule. Substitution shows that both equations change the provisional endpoint by $\eta(1 - p)A_\phi$. The native GAGA transitions and final observation noise are preserved. Every parent tensor, including its noise schedule, remains frozen. With the common 16-element embedding vocabulary, each head has 7,106 parameters; its hidden pair network is the same as the small FlowMol correction.

Physical supervision and selection. Each parent generates two trajectories for each of the same 128 TRAIN compositions, balanced across the four atom-count ranges in Appendix G. Two late field evaluations per trajectory give 512 provisional endpoints. FM records progress 0.734375 and 0.921875; GAGA records the nearest reverse-sampling evaluations, at progress $1 - 174/650$ and $1 - 51/650$. All states are retained. Mirror-paired eSEN queries provide centered forces, and the same bounded displacement rule gives the regression target $\delta(H)/(1 - p)$ for A_ϕ . No target requires capping by the per-atom velocity bound. Each head uses 20,000 AdamW updates on 96 TRAIN compositions; the other 32 measure target-prediction transfer. The four fits reduce this internal prediction error by 29.7–32.1% relative to zero correction.

Eight separate validation compositions select one strength per algorithm across both fits from $\{0, 1/4, 1/2, 1\}$. Selection maximizes graph-valid yield with GFN2 force RMS at most 5 eV/Å, including every attempt in the denominator; ties choose the smaller strength. Both algorithms

select $\eta = 1$. Only after this choice is audited and frozen do we generate the new test panel: 16 compositions, eight each with 17–28 and 29–40 atoms, with 16 outputs per composition and fit. None occurs in the prepared training or validation corpus, or in earlier generator evaluation panels.

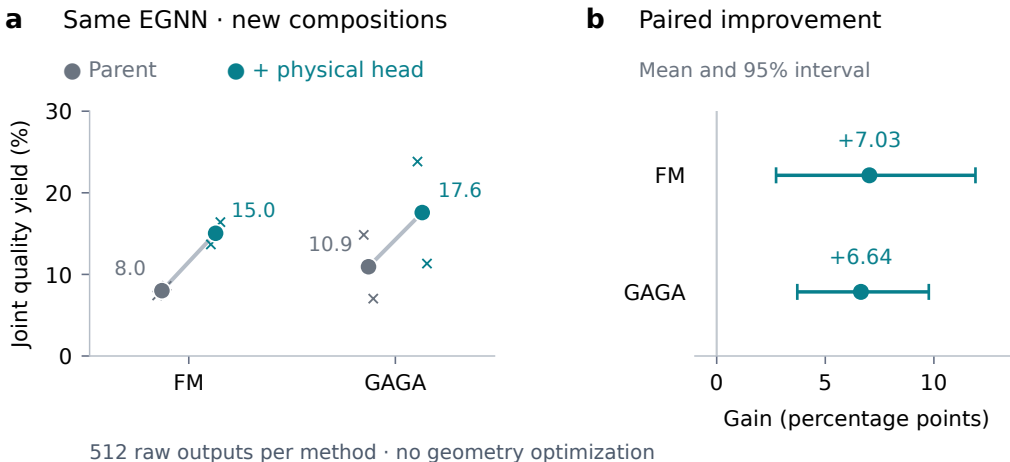


Figure 5: Physical correction transfers to independently trained EGNN generators. (a) Joint yield requires an inferred valid graph and GFN2 force RMS at most 5 eV/Å. Large circles show pooled rates; small crosses show the two fits. (b) Within-parent improvements with paired composition-bootstrap 95% intervals. Both algorithms receive the same TRAIN compositions, force-query budget, head architecture, and fitting schedule. All generated coordinates are unoptimized.

Table 15: Fresh shared-EGNN confirmation. Each row contains 512 attempts on the same 16 compositions. The last two columns give joint yield in each independent parent/head fit. Every GFN2 single-point calculation completed successfully.

Model	Graph (%)	Joint (%)	Fit 1 (%)	Fit 2 (%)
FM, distance feedback	17.58	8.01	7.42	8.59
FM + physical head	20.31	15.04	13.67	16.41
GAGA	18.75	10.94	14.84	7.03
GAGA + physical head	20.31	17.58	23.83	11.33

The primary FM comparison improves joint yield by 7.03 percentage points (95% interval [2.73, 11.91]), with gains of 6.25 and 7.81 points in the two fits. GAGA also benefits, gaining 6.64 points ([3.71, 9.77]), with both fits improving. Thus the learned correction transfers beyond the pretrained FlowMol backbone and works with both velocity and noise prediction.

After identical physical supervision, FM minus GAGA is -2.54 points ($[-6.64, 1.76]$). The difference between their within-parent improvements is $+0.39$ points ($[-4.30, 5.47]$). The experiment supports physical-head transfer to both generators, while their ordering remains uncertain. Intervals resample compositions and condition on the two fitted parents and heads; validation strengths are fixed before test generation.

Cost and verification. The study uses 1,024 TRAIN trajectories, 4,096 eSEN teacher queries, and 80,000 head updates in total. Validation and test each contribute 2,048 raw outputs and the same number of fixed-coordinate GFN2 calculations. Each output uses 128 backbone calls; FM uses 64 additional head calls and GAGA uses 128. The zero-strength controls execute the same head operations as their corrected counterparts. Supervision and head-update budgets match; total FLOPs and wall time are not equated. Generation uses no physical queries, geometry optimization, or energy-based selection. Zero heads and passive trajectory observers reproduce the native samplers exactly. Auditing reconstructs the targets, checks fitted head errors, re-evaluates all raw graphs, and independently parses xTB gradients.

H.1 CORRECTION STRENGTH AND THE QUALITY TRADEOFF

The validation study above selects the largest allowed strength for both generators. We extend the common grid to $\{0, 1/4, 1/2, 1, 2, 4\}$, reusing the same eight validation compositions and random seeds. Only strengths two and four require new samples. Parents, heads, force labels, and fitting budgets are unchanged. The applied correction bound scales with the strength η ; the bound stated for $\eta = 1$ is not retained at larger strengths.

We compare two validation choices. A conservative rule permits at most a one-point decrease in pooled graph validity and a three-point decrease in each fit, relative to strength one. It retains strength one for both generators. Selecting solely by joint yield chooses strength four for both, with validation joint yields of 31.64% for FM and 28.91% for GAGA. This second objective is declared after the conservative screen, before any new test generation. Both choices are then evaluated on the same 32 previously ungenerated compositions, 16 each with 17–28 and 29–40 atoms, with 16 outputs per composition and fit.

Table 16: Strength confirmation on the shared EGNN, with 1,024 unoptimized outputs per row. Force is the median RMS over each model’s graph-valid, successfully scored outputs. Sampling time is measured per output within the same GPU allocation for each fit, excluding checkpoint loading and physical assessment.

Model	Graph (%)	Joint (%)	Force (eV/Å)	Time (s)
FM, $\eta = 1$	20.12	15.14	3.833	0.123
FM, $\eta = 4$	22.17	20.80	3.159	0.122
GAGA, $\eta = 1$	20.80	16.50	3.805	0.143
GAGA, $\eta = 4$	21.00	20.02	3.158	0.143

Stronger correction improves FM joint yield by 5.66 percentage points (95% composition-bootstrap interval [2.73, 9.38]), with gains of 7.23 and 4.10 points in the two fits. The GAGA change is +3.52 points ([0.00, 7.52]). At strength four, FM minus GAGA is +0.78 points in joint yield ([−4.00, 6.54]) and +1.17 points in graph validity ([−2.93, 5.86]). The fit-specific joint differences are −1.37 and +2.93 points, so the pooled ordering is not stable across fitted models. The recorded FM sampling time is 14.6% lower, with 64 head calls per output versus GAGA’s 128; both use 128 backbone calls. These are measured sampling times, not equal training cost or a hardware-independent speed guarantee.

Across all outputs, stronger correction lowers FM’s mean GFN2 energy by 0.0622 eV/atom, with signed-change interval [−0.0769, −0.0479]. At strength four, FM’s mean energy remains 0.0527 eV/atom above GAGA’s ([0.0273, 0.0786]). These composition-paired readouts include invalid outputs and different inferred connectivities; they are not equilibrium free-energy differences. All graph-valid outputs within each 16-draw composition/fit cell have distinct canonical connectivities when stereochemistry is omitted.

The remaining errors are predominantly structural. At strength four, 533 of 1,024 FM outputs and 579 GAGA outputs fail the geometric connectedness test; only two FM outputs and no GAGA outputs overlap at the prescribed radius threshold, with both overlaps among the disconnected cases. Of the graph-valid outputs, 213/227 for FM and 205/215 for GAGA pass the force threshold. The extension adds 1,024 validation outputs and 4,096 confirmation outputs, each receiving a GFN2 attempt, with no new fitting or eSEN calls. One GAGA strength-one GFN2 calculation fails and remains in the denominator; the other 4,095 confirmation calculations succeed.

H.2 A CONDITIONAL FLOW FOR HYDROGEN COORDINATES

A connected heavy-atom skeleton can still fail molecular validation if a hydrogen lies too far from every heavy atom. In 512 cached training outputs per generator, FM produces 346 connected heavy skeletons and GAGA 314, but hydrogen-only fragmentation affects 60 FM outputs and 18 GAGA outputs. This motivates a conditional coordinate model that holds the heavy-atom geometry fixed and learns how the hydrogens should be arranged around it. Hydrogen decoration is also supported by existing generators such as Quetzal (Cheng et al., 2025); here we test it as a selective readout of a generated structure, with explicit geometric and physical controls.

Inputs, targets, and training. The model receives all current coordinates, atomic numbers, and a completion time $s \in [0, 1]$. Its output is a three-dimensional velocity for each hydrogen. For a training structure, we perturb hydrogen coordinates with Gaussian noise of standard deviation $0.4 \text{ \AA}$, keeping heavy atoms fixed. Half of the examples perturb every hydrogen; the other half perturb each hydrogen independently with probability one half. We pair perturbed and reference hydrogen positions by minimum squared-distance assignment, so the target does not depend on arbitrary labels of identical hydrogens. If Y_0 denotes the perturbed hydrogen coordinates and Y_1 their assigned reference positions, the training path and loss are

$$Y_s = (1 - s)Y_0 + sY_1, \quad \mathcal{L}_H = \mathbb{E} \left[\frac{1}{3N_H} \|u_\psi(X_{\text{heavy}}, Y_s, s; z) - (Y_1 - Y_0)\|_F^2 \right]. \quad (9)$$

Here X_{heavy} contains the fixed heavy-atom coordinates, z lists all atomic numbers, N_H is the hydrogen count, and u_ψ is the predicted hydrogen velocity. A common translation centers the full input; it does not change relative coordinates. Training uses coordinates and atom identities, without bond labels, energies, or forces.

We use a four-layer, width-64 EGNN with 153,790 total parameters, including 1,001 frozen schedule entries. Two independently initialized models each train for 10,000 updates of batch size 32 on the same 20,000 OMol25 structures used by the shared-backbone generators. The learning rate is $3 \cdot 10^{-4}$ and the exponential moving average decay is 0.999. We use the final averaged checkpoint. These fits add 640,000 training-example forward passes in total; the same two hydrogen models serve both FM and GAGA.

Generation and controls. The parent generators use their frozen strength-four physical corrections. A hydrogen activates the readout when its distance from every heavy atom exceeds 1.25 times their summed covalent radii. With no such hydrogen, the entire output is unchanged. Otherwise we integrate the conditional flow from $s = 0$ to 1, allowing all hydrogens in that molecule to move. Four midpoint steps require eight small-network evaluations per activated output. Each predicted hydrogen velocity is capped at $2 \text{ \AA}$ per unit completion time. Heavy atoms remain fixed during integration, and a final common translation restores the input centroid. Thus their relative geometry is preserved. No force or energy evaluator is called during this readout.

The fixed radial control moves each detached hydrogen toward its nearest heavy atom until their separation equals the sum of covalent radii. It leaves all other relative coordinates fixed. A second control applies the learned flow only to the hydrogens that triggered the readout, instead of moving all hydrogens together. Both controls start from exactly the same parent outputs. These are coordinate readouts, distinct from both the unchanged parent and force-field or quantum geometry optimization.

Selection and independent physical test. Sixteen development compositions compare the radial rule with the two learned modes, each starting at time zero or one half. Moving all hydrogens from time zero gives the highest FM joint yield, 96/256 versus the radial rule’s 95/256, but does not meet the requirement of improvement in both fits. Its lower energy motivates a separate physical hypothesis: relative to the radial rule, the learned readout lowers mean energy per atom while retaining pooled graph and joint yield. This hypothesis and the frozen model are specified before generation on 30 new compositions, 15 each with 17–28 and 29–40 atoms. Each fit draws 16 samples per composition, giving 960 attempts per setting. Composition bootstrap intervals retain both fitted models in every resample.

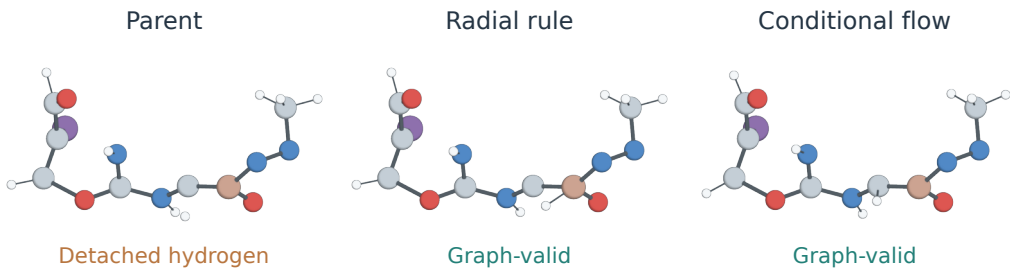
For FM, the learned readout lowers energy by 15.21 meV/atom relative to the radial rule (95% interval for the reduction $[10.10, 21.71]$), with reductions in both fits. Graph and joint yield each differ by only +0.10 percentage points ($[-0.21, 0.42]$). Relative to the unchanged parent, graph validity increases by 2.08 points ($[1.15, 3.13]$) and joint yield by 1.88 points ($[1.04, 2.71]$). Every previously graph-valid parent remains unchanged. Allowing all hydrogens to move, rather than only the detached ones, further reduces energy by 10.25 meV/atom ($[6.96, 14.72]$), with identical graph and joint counts. This ablation supports learning their arrangement together. GAGA also benefits physically: its learned readout lowers energy by 3.10 meV/atom relative to the radial rule ($[1.04, 6.34]$).

With the same learned readout for both generators, FM exceeds GAGA in graph validity by 3.33 points ($[0.52, 6.15]$). The joint difference is +2.29 points ($[-0.63, 5.21]$), so this physical-quality ordering

Table 17: Hydrogen readouts on 30 fresh compositions. All rows use the same strength-four physical heads. Joint yield additionally requires GFN2 force RMS at most 5 eV/Å. Energy is the all-output change relative to the corresponding unchanged parent, in meV/atom.

Parent	Hydrogen readout	Graph (%)	Joint (%)	Energy change
FM	None	25.31	24.17	0
FM	Radial rule	27.29	25.94	+1.89
FM	Conditional flow	27.40	26.04	−13.32
GAGA	None	23.85	23.54	0
GAGA	Radial rule	24.06	23.75	+1.05
GAGA	Conditional flow	24.06	23.75	−2.04

a Hydrogen placement around one fixed heavy skeleton



b Energy reduction relative to the radial rule

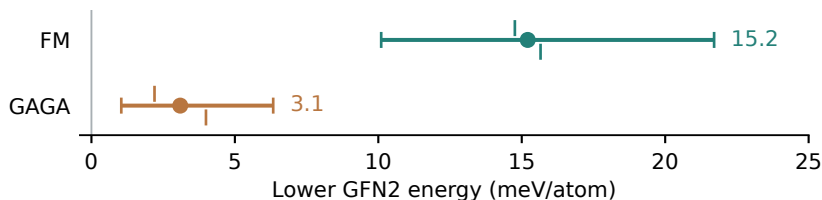


Figure 6: A conditional hydrogen flow learns more than attachment distance. Top: the same saved FM output before readout, after the radial rule, and after the learned flow. All atoms are shown with a shared camera and scale. Sticks are distance contacts in every panel, not inferred chemical bond orders. The example is selected by a fixed metadata hash among repaired, jointly graph-valid examples with at most 28 atoms; energy is not a selection criterion. Bottom: all-output energy reductions relative to the radial rule, with 95% composition-bootstrap intervals and the two fit-specific means. Carbon, hydrogen, nitrogen, oxygen, boron, and iodine are gray, white, blue, red, tan, and purple.

remains uncertain. The all-output energy averages include invalid geometries. As a descriptive check on valid outputs, 261 paired FM structures pass graph validation under both readouts; 18 are newly repaired. On these 18 pairs the learned readout has 83.81 meV/atom lower mean energy than the radial rule. This conditional subset is not used for model selection or for the primary energy interval.

The confirmation generates 1,920 parent trajectories and 5,760 derived outputs. It requires 2,460 new GFN2 calculations: one per parent and 540 unique changed geometries. Identical coordinates reuse the original calculation. These costs are separate from development, which uses 2,560 derived outputs and 155 new GFN2 calculations, and from fitting the hydrogen models. All sampled outputs remain in the reported denominators. The result establishes a local physical benefit of the readout, not sampling from a global Boltzmann distribution.

I COMPLETE GENERATORS ON COMMON COMPOSITIONS

We evaluate the complete systems on the same 16 new compositions, eight each with 17–28 and 29–40 atoms. The panel is fixed by metadata hashes, excludes prior confirmation panels, and is absent from both processed generator corpora. All conditions are neutral organic singlets within the GAGA adaptation’s element vocabulary. Each system generates 16 raw structures per composition and run, for 512 scored outputs per row of Table 18.

This comparison answers a different question from the shared-EGNN experiment. The FlowMol variants use a 5.90-million-parameter inherited backbone; the GAGA models use the 2.38-million-parameter EGNN trained from scratch on 20,000 OMol25 structures. Their architectures, training histories, physical supervision, and costs differ. GAGA uses its validation-selected truncation at 650, evaluated at both 128 calls and the complete 651-call schedule. The paired-update model keeps its two original continuations. The correction-head study uses two head-fitting seeds on one shared parent. Intervals condition on these fitted systems and resample the 16 compositions.

Same compositions · raw generated coordinates

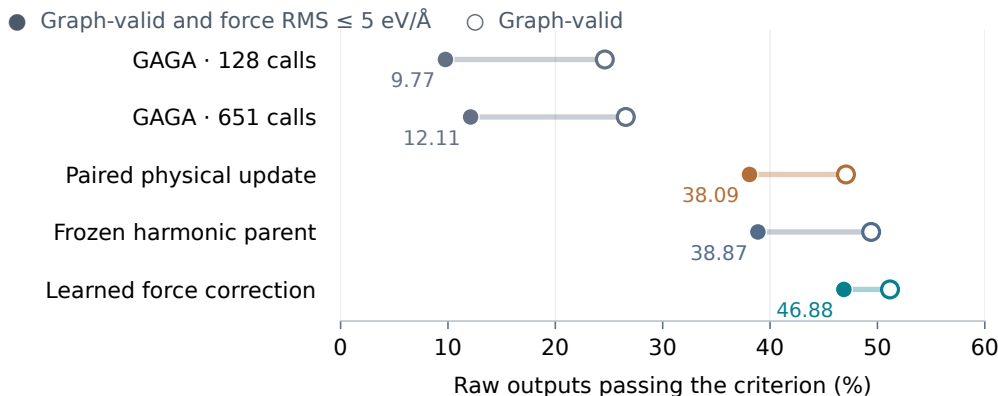


Figure 7: Graph support and physical quality on the same compositions. Open circles count graph-valid outputs; filled circles also require GFN2 force RMS at most 5 eV/Å. The gap shows graph-valid outputs above that force threshold. Every denominator includes all 512 attempts. The models have the distinct training histories and inference costs described in the text.

Table 18: Raw outputs in the common-panel study. Force medians use each model’s own graph-valid outputs with successful GFN2 calculations. Joint yield uses all attempts. “Calls” separates backbone and correction-head evaluations.

Model	Calls	Graph (%)	Force (eV/Å)	Joint (%)
GAGA	128	24.61	5.408	9.77
GAGA, complete schedule	651	26.56	5.176	12.11
Paired physical update	128	47.07	3.581	38.09
Paired update, no terminal noise	128	46.88	2.929	40.04
Frozen harmonic parent	128+64	49.41	3.394	38.87
Small correction, 2,000 updates	128+64	50.00	3.077	43.16
Small correction, 20,000 updates	128+64	51.17	2.854	46.88
Wide correction, 20,000 updates	128+64	50.20	2.831	46.09
Contextual correction, 20,000 updates	128+64	50.78	2.864	46.68

The contextual head was the prespecified primary challenger. It improves joint yield over its parent by 7.81 percentage points, with a 95% interval of [4.88, 10.94]; the run-specific gains are 7.03 and 8.59 points. Its differences from the small and wide 20,000-update heads are -0.20 points ($[-1.95, 1.56]$) and $+0.59$ points ($[-1.17, 2.34]$). Thus the better internal force-target prediction of the contextual architecture does not produce an established generation advantage over either simpler control.

Relative to GAGA at 128 and 651 calls, the contextual complete system gains 36.91 points ([26.56, 46.09]) and 34.57 points ([24.80, 43.95]). The original paired-update system also exceeds both GAGA settings, by 28.32 points ([20.70, 35.74]) and 25.98 points ([18.75, 33.40]). These are complete-system results on common conditions. The shared-backbone results in Appendices F.1 and H control architecture and training exposure. Learned correction improves both EGNN generators under equal physical supervision, with uncertain ordering between them.

Training duration and terminal noise. The published-model and terminal-noise controls were designated before reading the new GFN2 comparison. The zero-noise output is captured from the same neural trajectory as the published sampler, before its final 0.025 Å perturbation. Removing that noise increases joint yield by 1.95 points ([0.39, 3.71]). The small learned correction exceeds the paired-update model by 8.79 points ([3.91, 13.67]), and its zero-noise version by 6.84 points ([1.95, 11.72]). Their physical-training sets and parent histories differ, so this comparison combines those choices with the adaptation mechanism.

After inspecting the main confirmation, we add a training-duration ablation using the already-saved 2,000-update prefix of each small-head fit. All targets, initial weights, training-state choices, and generation settings match the 20,000-update model. Longer fitting adds 3.71 points ([1.76, 5.86]), with positive changes in both runs. This supplemental ablation is a later diagnostic, not an independently chosen confirmation panel.

Across all unoptimized outputs, the small 20,000-update head also lowers GFN2 energy by a mean 0.0951 eV/atom relative to its matched parent. The signed-change 95% interval is $[-0.2022, -0.0364]$ eV/atom, and both runs decrease. This readout includes invalid outputs and measures the change across paired generator attempts, rather than equilibrium energy differences between isomers.

Additional graph diagnostics. Requiring RDKit to assign zero radical electrons gives joint yields of 46.29% for the small correction, 46.09% for the contextual correction, and 9.57% and 11.52% for the two GAGA budgets. The relative result therefore also appears under this stricter graph-assignment check. Within each composition and run, all graph-valid outputs have distinct inferred labeled connectivity graphs. This 16-draw diagnostic measures observed duplication, not coverage of the possible isomers or their equilibrium probabilities.

The six-system confirmation contributes 3,072 neural trajectories and GFN2 readouts. Its parameter-matched control adds 40,000 head updates; the preceding small/contextual capacity study used 80,000 updates. The supplemental published and short-fit study adds 1,024 neural trajectories and 512 further outputs obtained by sharing the published model’s trajectory across the two noise settings, with 1,536 GFN2 readouts. The physical teacher is reused throughout. No generated coordinates are optimized.

I.1 LEARNING FROM PHYSICALLY CORRECTED REFERENCES

We test a second way to supply physical training targets: correcting OMol25 reference geometries before fine-tuning. This separates the value of physical labels from the effect of additional training on a small reference set.

Matched reference sets. From the existing 3,000-row training block, we select 128 distinct compositions by fixed metadata hashes, with 32 in each of the 8–16, 17–24, 25–32, and 33–40 atom ranges. Each contributes one reference structure. We construct a corrected copy using a bounded Cartesian L-BFGS search under the inversion-averaged eSEN potential. All coordinates may move, while geometric validity and the inferred labeled graph are preserved. Each search allows at most 96 steps and 160 energy/force evaluations, with a 0.05 Å per-atom step cap. It stops early when the largest atomic force is at most 0.1 eV/Å.

Every composition remains in both sets. If a search stalls or reaches its budget, we retain its last accepted point and record its termination status. Of the 128 searches, 110 meet the force stopping criterion and 124 lower force RMS. Including inversion pairs, target preparation uses 6,576 eSEN evaluations. Independent GFN2 single points evaluate all 256 original and corrected structures.

Table 19: Quality of the 128 training references before and after physical correction. These are training labels, not generated outputs. GFN2 scores are used only for evaluation.

Metric	Original references	Corrected references
Median eSEN force RMS (eV/Å)	2.005	0.050
Median GFN2 force RMS (eV/Å)	1.756	0.397
GFN2 force RMS ≤ 1 eV/Å (%)	22.66	89.84

Direct flow-matching adaptation. Two fine-tuning runs start from the same 5.90-million-parameter harmonic-source checkpoint. Within each run, one student learns from the original references and another from their corrected copies. Each receives 2,000 updates at batch size one. They share reference indices, source draws, rotation and atom-matching randomness, optimizer settings, and 0.005 Å centered endpoint smoothing. Composition frequencies are uniform and identical in the two target sets. This experiment uses direct FM training without parameter subtraction; its target is the specified empirical reference distribution.

The confirmation panel contains 16 previously ungenerated compositions, eight each with 17–28 and 29–40 atoms, absent from both processed corpora and earlier generation panels. Each model generates 16 samples per composition and run. All three models use 128 backbone calls and zero terminal noise. The original parent is regenerated with the same settings. No checkpoint is selected using these outcomes, and no generated structure is physically optimized.

Table 20: Raw generation on the reference-target confirmation panel, with 512 attempts per model. GFN2 joint yield requires chemical-graph validity and force RMS at most 5 eV/Å, divided by all attempts. Force medians use each model’s own graph-valid outputs with successful calculations.

Model	Graph validity (%)	Median force (eV/Å)	Joint yield (%)
Unadapted parent	38.09	3.083	31.64
Original-reference fine-tuning	30.47	2.762	27.34
Corrected-reference fine-tuning	39.45	2.558	33.01

Corrected references improve joint yield over matched original-reference fine-tuning by 5.66 percentage points, with a composition-bootstrap 95% interval of [2.54, 8.59]. The run-specific gains are +8.20 and +3.13 points. Relative to the unadapted parent, the pooled gain is smaller: +1.37 points ([−3.52, 6.05]), with run-specific changes of −1.95 and +4.69 points. Thus physical labels help compared with the same additional reference training, while an overall advantage over the parent remains uncertain. The intervals condition on these two fitted runs. This study adds 8,000 optimizer updates, 1,536 raw neural outputs, and 1,536 GFN2 generation readouts, in addition to target preparation and its 256 GFN2 label checks.

Physical evaluation of the frozen source models. This evaluation concerns the pretrained source models on the 10-composition additional test set. We score every generated output and its original reference with the same conservative eSEN checkpoint at the original charge and spin. Energies and forces are averaged over inversions as $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. We pair differences within composition before averaging. Per-atom energies also subtract the composition’s reference energy; that reference cancels in a paired model difference. The evaluation uses 5,160 raw energy/force queries, including both inversions of references in each fine-tuning run.

For each sample, force RMS is the square root of the mean squared atomic force norm; we then average this value over samples. The jointly valid comparison retains only pairs accepted under both models, with 3–35 pairs in each composition/run combination. The all-attempt comparison also includes graph-invalid outputs. Reporting both distinguishes overall scores from scores on pairs that remain valid under both methods.

Most of the aggregate improvement comes from run 1. The all-attempt energy differences are −0.08049 and −0.00070 eV/atom in runs 1 and 2, respectively; the force-RMS differences are −1.21996 and +0.29302 eV/Å.

Table 21: Harmonic-minus-Gaussian eSEN scores on the frozen models’ 10-composition test. Negative values favor harmonic. Rows distinguish all attempts from paired outputs that are both graph-valid. Brackets give paired 95% confidence intervals.

Metric	Outputs	Difference	Interval
Energy (eV/atom)	All attempts	−0.0406	[−0.0579, −0.0232]
	Both outputs valid	−0.0340	[−0.0692, −0.0003]
Force RMS (eV/Å)	All attempts	−0.4635	[−0.6782, −0.2512]
	Both outputs valid	−0.5389	[−0.8950, −0.2089]

The original endpoint checkpoint. We also evaluate the original 30,000-step OMol25 checkpoint on coordinate generation, fixing the composition and neutral/no-bond inputs. Its endpoint sampler uses a first-step bootstrap and 127 Euler updates, for 128 network calls. Keeping the model’s predicted categorical history gives 106/640 and 113/640 graph-valid outputs in two sampling streams. Fixing the history categories and centering the initial bootstrap gives 116/640 and 120/640. Both streams use the same checkpoint; the second variant changes both history handling and bootstrap centering. LatentSpring’s source models additionally receive shared task adaptation and 3,000 specialized updates. The matched Gaussian and covariance-Gaussian fine-tuning runs therefore provide the controlled tests of source choice.

J NORMALIZATION OF THE HARMONIC MIXTURE

We first compute the density for one tree, then average over trees. The key is the change of coordinates from independent edge vectors to centered atom positions. For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be its edge-incidence matrix. Each row has $+1$ and -1 at the endpoints of one edge, so $B_T X$ contains the edge displacements. Let $U \in \mathbb{R}^{N \times (N-1)}$ have orthonormal columns that span the centered subspace for one spatial dimension.

Because $B_T \mathbf{1} = 0$, we have $(B_T U)(B_T U)^\top = B_T B_T^\top$. By Cauchy–Binet, $\det(B_T B_T^\top)$ is the sum of the squares of its N maximal minors. Every maximal minor of a tree incidence matrix is ± 1 , so this determinant is N . The absolute Jacobian is therefore $\sqrt{N}$ in one spatial dimension and $N^{3/2}$ in three dimensions. The density for a fixed tree is consequently $q(X \mid T, c) = N^{3/2} \prod_{e \in T} h_e(B_T X)$, with each edge density evaluated at its corresponding row of $B_T X$. It integrates to one. Averaging these normalized components with Equation (1) gives Equation (3) and preserves normalization.

K SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Alignment chooses a rotation and atom relabeling using the reference structure. We show why a subsequent shared random augmentation restores the source marginal, even when the source is not Gaussian.

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Assume that q_0 is invariant under G and $X \sim q_0$. The pairing procedure selects $g(X, Y) \in G$ using the target Y . Independently draw H from the uniform, or Haar, distribution on G . Conditional on the orbit of X and the selected g , the transformed source HgX has the same uniform orbit distribution as HX , by Haar invariance. An orbit here is the set of configurations obtained by applying these symmetries to one configuration. Alignment changes neither the orbit nor its probability. Averaging over source orbits therefore recovers q_0 .

The paired endpoints can still be correlated. The training objective uses their conditional displacement target directly; this argument establishes the source marginal, not independence of the endpoints.

L COVARIANCE CONTROLS

The shell-tree control uses the same radius-based edge scale but concentrates edge lengths around it. For each edge, it samples a length $R = \ell \exp(sZ - s^2/2)$, where $Z \sim N(0, 1)$ and $s = 0.2$, then

samples a uniform direction. Since $\mathbb{E}R^2 = \ell^2 \exp(s^2)$ and the direction has second moment $I/3$, the edge covariance is $vI = \ell^2 \exp(s^2)I/3$.

Let A_T map edge coordinates to centered atom coordinates. Independence of edges gives $C_T = A_T \text{diag}(v_e) A_T^\top$, where C_T is the atom-position covariance for one spatial axis and the diagonal matrix contains edge variances. Replacing the shell vectors by independent Gaussians with covariance $v_e I$ leaves C_T unchanged. All tree components have zero mean, so the mixture covariance $\mathbb{E}_T C_T$ is also unchanged.

The single-Gaussian control estimates this mixture covariance by averaging a finite number of tree covariances, with randomness seeded by composition. We then average over permutations within each atom species. This preserves positive definiteness on the centered subspace and makes the covariance invariant to relabeling identical atoms. The resulting Gaussian is exactly normalized for its chosen covariance, but the covariance only approximates that of the full tree mixture.

M JARZYNSKI IDENTITY FOR ESCORTED TRANSPORT

This appendix states the change-of-variables identity behind our work-corrected local targets. Jarzynski’s relation connects exponential work averages to free-energy differences, including in single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport adds an imposed coordinate flow and accounts for its volume change (Vaikuntanathan & Jarzynski, 2008). Here, “work” denotes the corrected energy-like quantity whose exponential supplies an importance weight.

Fix a composition and use volume measure on its centered coordinate space. Let x collect the independent coordinates and let $\beta = 1/(k_B T_{\text{phys}})$. The unnormalized target is $\rho_1(x) = \mathbf{1}_\Omega(x) \exp[-\beta E(x)]$, with finite, nonzero $Z_1 = \int_{\mathcal{H}_N} \rho_1(x) dx$. The allowed region Ω can, for example, confine the molecule so that this integral is finite. The indicator is one inside this region and zero outside it; the normalized target is ρ_1/Z_1 .

Let $q_0 > 0$ be the normalized source in Equation (3) and let $\Phi : \mathcal{H}_N \rightarrow \mathcal{H}_N$ be a differentiable, invertible map with nonzero intrinsic Jacobian. Its absolute determinant $|\det D\Phi|$ measures local expansion or compression in the independent coordinates. For $\Phi(x_0) \in \Omega$, define

$$W_\Phi(x_0) = E(\Phi(x_0)) + \beta^{-1} \log q_0(x_0) - \beta^{-1} \log |\det D\Phi(x_0)|, \quad (10)$$

and set $W_\Phi = +\infty$ outside the allowed region, giving zero weight there. The source-density and Jacobian terms correct for where the mapped samples came from. Multiplying the exponential weight by the source density and changing variables gives

$$\mathbb{E}_{q_0}[e^{-\beta W_\Phi}] = \int_{\mathcal{H}_N} \rho_1(\Phi(x_0)) |\det D\Phi(x_0)| dx_0 \quad (11)$$

$$= Z_1. \quad (12)$$

Equivalently, the auxiliary source energy $E_0 = -\beta^{-1} \log q_0$ has $Z_0 = 1$ in this measure, yielding a Jarzynski-type normalizer identity. For an observable f , we estimate the target mean with $\sum_j w_j f(\Phi(x_j)) / \sum_j w_j$, where $w_j = e^{-\beta W_\Phi(x_j)}$. This is self-normalized importance sampling. Its finite-sample variance depends on how well the mapped proposals cover the target.

For an exact, sufficiently regular ordinary differential equation (ODE) flow, the intrinsic log determinant equals the integrated velocity divergence. Our implemented work corrections instead use explicit maps with tractable local densities: Gaussian translations in Appendix N and periodic rotor maps with analytic Jacobians in Appendix O. We do not use an exact neural-flow likelihood to evaluate the reported generator. Its outputs use the midpoint solver and terminal noise in Section 4.

N LOCAL FORCE-ESCORTED WORK DISTILLATION

The physical teacher constructs nearby candidate structures and determines how often each is used as a flow-matching target. We compare uniform sampling of force-shifted candidates with energy-weighted training (Dern et al., 2025; Midgley et al., 2023). A normalized Gaussian around each anchor makes the complete escorted-work weights explicit (Vaikuntanathan & Jarzynski, 2008).

Energy, force, and mirror symmetry. The energy model gives a scalar $E(X)$ and forces $F(X) = -\nabla_X E(X)$. Each atom’s force points toward decreasing energy. To treat mirror-related structures consistently, we evaluate both X and its inversion $-X$ and use $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. The minus sign in the force average ensures $F_+ = -\nabla E_+$. Inverted structures thus receive equal energies and opposite forces. Before constructing a displacement, we subtract the mean atom force so the displacement stays in the centered coordinate space.

Local reference and target. For a graph-valid generated anchor A , let G_A be its inferred labeled graph and define $q_A(X) = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. We use thermal energy $kT = k_B T_{\text{phys}} = 0.0258519998$ eV at 300 K and set

$$\sigma_A = \min \left(0.03 \text{ \AA}, \sqrt{\frac{0.10 \text{ \AA} kT}{\|F_+(A)\|}} \right), \quad \delta_A = \frac{\sigma_A^2}{kT} F_+(A). \quad (13)$$

The norm is the Frobenius norm across all atom-force components. These choices bound the displacement norm by 0.10 Å and the Gaussian width by 0.03 Å. At zero force, we use the maximum width. Both quantities are fixed before sampling $X \sim q_A$. We then translate each candidate to $Y = X + \delta_A$. This map stays within the centered subspace and has determinant one.

The unnormalized local target is

$$\rho_A(Y) = q_A(Y) \exp \left[-\frac{E_+(Y) - E_+(A)}{kT} \right] \mathbf{1}_{G_A}(Y). \quad (14)$$

The Gaussian keeps structures near the anchor, while the energy factor favors lower-energy structures. The indicator restricts candidates to the anchor’s inferred labeled bond orders, formal charges, and radical counts, together with the connected/nonoverlap checks. Thus this is a restrained local distribution, not an unrestricted equilibrium distribution over molecules.

Why the force displacement has this form. Approximate energy near the anchor by $E_+(Y) - E_+(A) \simeq -\langle F_+(A), Y - A \rangle_F$. The inner product sums force times displacement over all atoms and coordinates. Multiplying this linearized Boltzmann factor by the Gaussian and completing the square gives

$$q_A(Y) \exp \left[\frac{\langle F_+(A), Y - A \rangle_F}{kT} \right] \propto \mathcal{N}_{\mathcal{H}_N} \left(Y; A + \frac{\sigma_A^2}{kT} F_+(A), \sigma_A^2 I \right). \quad (15)$$

Before applying the graph restriction, the force translation therefore samples the energy-weighted Gaussian exactly under the linear approximation. Equivalently, the Gaussian restraint is a virtual harmonic spring with stiffness kT/σ_A^2 , and the force shifts its mean by force divided by stiffness. The same-graph check then truncates this distribution.

Complete local work. Uniform weighting uses the linear approximation above. Complete-work weighting instead evaluates the nonlinear energy to correct departures from that approximation. With reference Hamiltonian $H_0 = -kT \log q_A$ and target Hamiltonian $H_1 = H_0 + E_+ - E_+(A)$ plus the hard support restriction, the work is

$$W_A(X) = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)] \quad (16)$$

on allowed candidates, and $+\infty$ otherwise. The log-density terms correct for the shift from the proposal X to Y . Change of variables gives $\mathbb{E}_{q_A}[\exp(-W_A/kT)] = \int \rho_A(Y) dY$. Analytic checks recover constant work for a constant-force translation, and the normalizer and weighted mean for a quadratic energy. Substituting the linearized energy in Equation (15) gives $W_A = -\sigma_A^2 \|F_+(A)\|_F^2 / (2kT)$ for every allowed candidate. Their weights are therefore equal in that approximation, explaining the less expensive force-only teacher.

Training pools and controls. Each fine-tuning run selects eight distinct training compositions with 17–28 atoms using a fixed metadata hash, then generates 32 anchors per composition. These compositions are excluded from evaluation. Invalid anchors count as generation attempts but do not enter the teacher pool. Each valid anchor produces eight force-shifted candidates; candidates outside its support receive zero weight. The complete-work model samples training endpoints with

Table 22: Graph-valid outputs after direct physical fine-tuning, out of 640 attempts per run. The frozen model receives no updates; all other variants have the same additional training budget.

Run	Frozen	Replay	Unweighted escort	Complete work
1	418	340	330	344
2	377	345	371	362

self-normalized $\exp(-W_A/kT)$ weights. The unweighted-escort model samples uniformly from the same allowed candidates. The replay model uses the original retained anchors, while the frozen model receives no further training. Anchors with no allowed candidate are recorded and excluded from all student pools in the same way.

Each student alternates between 500 FM updates on original references and 500 updates on generated endpoints. Initialization, anchor draws, optimization, and sampling budgets are shared. Replay needs no force queries; the unweighted escort needs anchor forces; complete work additionally needs energies at valid candidate endpoints. We distinguish the queries used in the shared research experiment from those required by each method on its own. Generation uses only the final network. Its temperature feature is fixed during training and evaluation; 300 K sets the local teacher weights.

Self-normalized weights introduce finite-particle bias. We report effective sample size (ESS) over each anchor’s eight candidates: $(\sum_j w_j)^2 / \sum_j w_j^2$. Equal weights give ESS equal to the candidate count, while a single dominant candidate gives ESS close to one.

Direct fine-tuning results. We train three students for 1,000 updates in each run. Across 512 generated anchors, 351 pass the graph checks and 349 have at least one allowed force-shifted candidate. Composition-level mean ESS is about 1.27–1.87 out of 8. The weights are therefore concentrated on few candidates, limiting resolution of the local target with this pool size.

Among paired outputs that are both graph-valid, complete-work training gives lower energy than unweighted-escort training by 0.00758 eV/atom (paired 95% interval $[-0.01424, -0.00113]$ for the signed difference). Force RMS is lower by 0.16174 eV/Å ($[-0.27660, -0.05590]$). Both runs favor complete work, and the all-output comparison has the same direction.

However, all three directly fine-tuned students perform worse than the frozen model. Complete work has 55.16% graph validity versus 62.11% for the frozen model, and energy on paired, jointly valid outputs is *higher* by 0.10587 eV/atom ($[0.09009, 0.12168]$). Replay also worsens performance after training on the mixed targets. These results motivate subtracting the matched replay update rather than using the physical student directly.

Teacher preparation uses 6,232 raw energy/force queries: 702 at anchors and 5,530 at allowed candidates. The unweighted escort could omit candidate-energy queries, and replay could omit all physical queries. Scoring the outputs of the four models uses a further 10,240 queries, separate from generation cost.

Applying the paired physical update. Instead of deploying the directly fine-tuned student, we use parameter arithmetic (Ilharco et al., 2023) to subtract its matched replay update:

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}}), \quad (17)$$

for both unweighted-escort and complete-work students. The coefficient is fixed at one. We change parameter tensors only; buffers and the source distribution remain fixed.

Both paired variants lower energy and force RMS in both runs. For complete work, force RMS on paired, jointly valid outputs falls by 0.38403 eV/Å (interval $[-0.48124, -0.28932]$); all-attempt energy and force scores also improve. Its graph-validity difference from the frozen model is -0.55 percentage points, with interval $[-2.58, 1.56]$.

The complete-work and unweighted paired variants have similar scores. Complete work minus unweighted gives -0.00243 eV/atom on jointly valid pairs ($[-0.00756, 0.00264]$) and $+0.00022$ eV/atom over all attempts ($[-0.00557, 0.00649]$). Force and graph-validity intervals also include zero. Teacher preparation requires 702 anchor queries for the unweighted variant versus 6,232 for complete work. Both retain the 128-call neural sampler.

Table 23: Paired updates evaluated after the direct fine-tuning experiment. Energy changes use paired, jointly valid outputs and are relative to the frozen model, with paired 95% confidence intervals. Counts show the two generation streams.

Model	Graph-valid outputs (two streams)	Energy change (eV/atom)
Frozen	418/640; 377/640	0
Paired force update	407/640; 387/640	-0.01554 [-0.02084, -0.01031]
Paired work update	406/640; 382/640	-0.01604 [-0.02188, -0.01013]

Evaluating these two paired variants adds 2,560 generated outputs and 5,120 physical queries. Combining parameters requires no optimizer steps. Based on these development results, we select the force-only update for the new-composition evaluation in Section 6.4.

O MOLECULAR ROTOR WORK PROTOCOL

We use a controlled one-dimensional motion to test the work correction on molecular energy curves. The three training molecules are C#C(N)N(C)C(N)=O, CCOC(C)(C)N, and CC(=O)NCC=O. We inspect the first five rows of an existing training subset and take the first three distinct compositions with a methyl rotor that preserves the inferred labeled graph on a 64-angle grid. Selection uses no energy queries. Only the methyl hydrogens rotate around the carbon-anchor axis; other internal coordinates remain fixed. We remove overall translation and define densities with respect to the periodic angle measure $d\phi$ on $[-\pi, \pi)$.

Reference, target, and coordinate map. The starting density is $q_0(\phi) = \exp(\cos \phi)/(2\pi I_0(1))$, with normalizing integral $2\pi I_0(1) = \int_{-\pi}^{\pi} \exp(\cos u) du$. This is a circular reference distribution, distinct from the coordinate source used by the neural generator. Its reference Hamiltonian is $H_0 = -kT \log q_0$, and the target Hamiltonian is $H_1 = H_0 + U - U(0)$. We transform angles with $\psi_a(\phi) = \phi + a \sin \phi$ for $a \in \{-0.5, 0, 0.5\}$. The derivative, $J_a(\phi) = 1 + a \cos \phi$, is the local volume-change factor. Since $|a| < 1$, this map is smooth and invertible on the circle. Its complete work is

$$W_a(\phi) = U(\psi_a(\phi)) - U(0) + kT [\log q_0(\phi) - \log q_0(\psi_a(\phi)) - \log J_a(\phi)]. \quad (18)$$

The density terms correct the changed angular sampling distribution, and the Jacobian term corrects expansion or compression of angular intervals. Changing variables gives

$$\mathbb{E}_{q_0} e^{-W_a/kT} = \int_{-\pi}^{\pi} q_0(y) e^{-[U(y)-U(0)]/kT} dy = Z_1/Z_0, \quad Z_0 = 1. \quad (19)$$

Numerical reference. All 1,536 single-point calculations succeed. For each rotor, we construct a periodic cubic interpolation of 512 GFN2 energy values. The 256-knot interpolation has error at most 10^{-8} eV at the additional midpoint samples in this study. Increasing the reference quadrature from 32,768 to 65,536 points changes free energy by less than 10^{-10} eV. These checks assess interpolation and numerical integration errors for the GFN2-based reference curve.

Estimators and finite-sample effects. We compute total variation on 64 equal angular bins, using dense quadrature for the target probabilities. Self-normalized estimates of observables have finite-sample bias, whereas the average unnormalized weight is an unbiased estimator of the normalizing constant. The free-energy estimate is $-kT \log \hat{Z}$; its RMSE includes the bias introduced at finite sample size. At 2,048 draws, complete-work ESS ranges from 788 to 1,892 across the six nonidentity cases. Some incomplete weights have higher ESS despite targeting the wrong distribution. ESS alone therefore does not establish correct target recovery.

O.1 COMPLETE-WORK CORRECTION ON MOLECULAR ROTORS

For this test, one angle ϕ specifies a family of three-dimensional structures: the three hydrogens of a methyl group rotate around its bond axis while the other internal coordinates remain fixed.

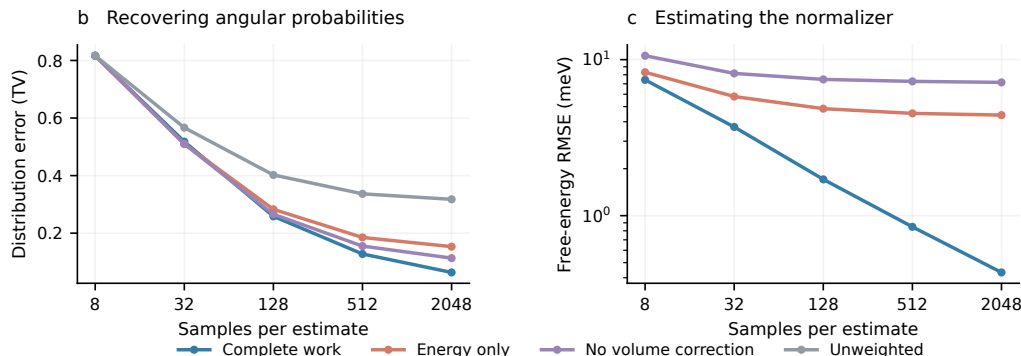
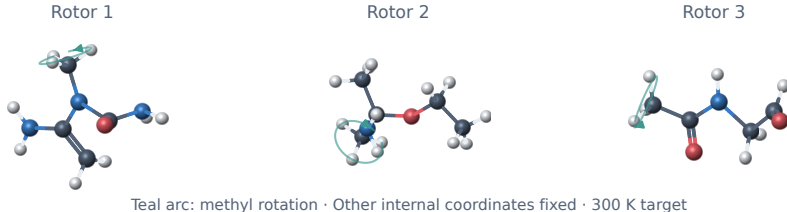
a Controlled methyl rotation on three molecules

Figure 8: Work correction on controlled molecular rotors at 300 K. (a) Reference structures and the prescribed methyl rotations. (b,c) Angular-distribution and normalizer errors averaged over three molecules and two nonidentity maps, with 128 repetitions per case. Complete weights account for the proposal bias left by energy-only and incomplete weights. Unweighted samples have no free-energy estimator. Each target retains its angular reference restraint.

We select three such rotors from training molecules in fixed data order using geometry alone. A periodic interpolation of 512 GFN2 energies defines $U(\phi)$ for each rotor, with $U(0)$ as the reference energy. The target is $\rho(\phi) \propto q_0(\phi) \exp[-(U(\phi) - U(0))/kT]$, where q_0 is a normalized von Mises distribution with concentration one. It favors angles near zero; the energy factor reweights those angles while retaining this restraint. Here $kT = k_B T_{\text{phys}}$ at 300 K.

We apply three fixed angle maps, including the identity map, to the starting samples. For each map, we compare complete-work weighting, energy-only weighting, weighting without the Jacobian correction, and no weighting. Complete work accounts for both the reference-density change and the expansion or compression of angle intervals. Each setting uses 128 repetitions at sample counts from 8 to 2,048. The maps and work formula are specified above in Appendix O.

Total variation measures the discrepancy between estimated and target angular probabilities; zero indicates agreement. Free-energy RMSE measures error in the logarithm of the target normalizer, expressed in energy units.

At 2,048 samples, complete work has lower mean total-variation error than both incomplete-weight controls in all six nonidentity cases. The average error is 0.0636 for complete work, 0.1532 for energy-only weights, and 0.1134 without the Jacobian. Complete work therefore reduces this error by 58.5% relative to energy-only weighting. Its free-energy RMSE ranges from 0.158 to 0.734 meV. Independent density-ratio quadrature also recovers the target normalizer in all nine molecule/map cases. With the identity map, the three weighted methods coincide. These results test the full work correction on restrained molecular energy curves, rather than equilibrium sampling by the neural generator.

P PHYSICAL QUALITY OF THE MATCHED GENERATORS

We evaluate physical scores for the 16,384 saved outputs from the independent EGNN comparison in Section 6.3. These are the original 128-call outputs; no models are retrained and no coordinates are changed. We fix the physical-evaluation protocol before computing these scores. eSEN evaluates every output and its reflection. GFN2 evaluates every graph-valid output and both inversions of each reference. All 2,198 GFN2 calculations succeed, including 256 reference checks that pass inversion tests. eSEN uses 33,024 evaluations.

Joint quality yield divides outputs that pass both graph validity and a physical threshold by *all generation attempts*. Graph-invalid structures therefore remain in the denominator. Figure 9 sweeps the force-RMS threshold instead of treating one cutoff as a universal definition of physical validity. Passing a threshold does not certify a stationary geometry. These models share the architecture and training budget in Appendix F; none receives a physical update in this source/objective comparison.

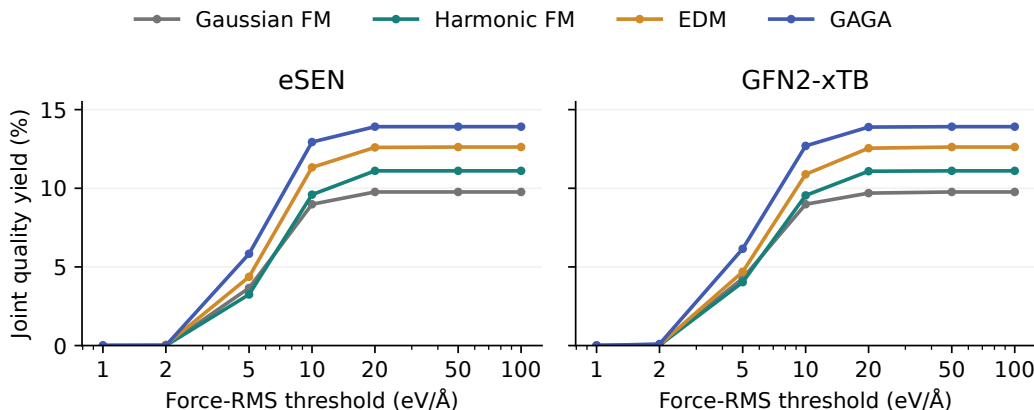


Figure 9: Fraction of all attempts that are graph-valid and have force RMS below the indicated threshold. Curves pool both independent initializations and all 64 compositions. GAGA has the highest pooled yield at most nontrivial thresholds under both potentials.

Table 24: GFN2 scores for the matched generators. Energy and force are medians across each method’s own graph-valid outputs; yield uses all 4,096 attempts per method. Energy excess is relative to one OMol25 geometry of the same composition. The generated bond graph may differ from that reference, so this is not a same-graph strain-energy measurement.

Method	Energy excess (eV/atom)	Force RMS (eV/Å)	Yield at 5 eV/Å (%)
Gaussian FM	0.6820	5.2911	4.27
Harmonic FM	0.7114	5.8183	4.03
EDM	0.6799	5.8768	4.69
GAGA	0.6604	5.2785	6.15

Higher graph validity from the harmonic source does not, by itself, imply better physical scores among valid outputs. GAGA has stronger pooled physical results in this matched setting. The separate physical-adaptation experiments test the effect of local physical targets on trained generators.

Q TRAINING-COMPONENT CONTROLS

Q.1 WHAT SELF-CONDITIONING CONTRIBUTES

We remove self-conditioning from the harmonic-source model while keeping its initialization, 3,000 training structures, source distribution, and pairing procedure unchanged. One single-pass model receives 3,000 optimizer updates. A second receives 6,000, matching the self-conditioned model’s

Table 25: Self-conditioning ablation on 24 compositions. Each run has 768 raw attempts; the valid-output column lists runs 1 and 2. The self-conditioned model outperforms both single-pass training budgets.

Model	Updates	Training passes	Valid outputs	Rate (%)
Single pass	3,000	3,000	363;336	45.51
Single pass	6,000	6,000	277;342	40.30
Self-conditioned	3,000	6,000	435;375	52.73

number of backbone training evaluations. Both single-pass models use 64 midpoint steps at generation, so every model uses 128 network calls per output.

Self-conditioning improves graph validity by 7.23 percentage points at equal optimizer updates (95% interval [4.69,9.77]) and by 12.43 points at equal training forward passes ([9.70,15.23]). Both runs improve in both comparisons. Matching pass counts does not match every operation or wall-clock cost. This control asks whether additional single-pass training recovers the validity gain; it does not in these runs. The update-direction control is reported in Section 6.5.

Contribution of the force displacement. We also remove only the force shift from local training targets. Gaussian perturbations, candidate indices, widths, training schedule, and the paired parameter update remain unchanged. Relative to this control, force-shifted targets lower energy on paired, jointly valid outputs by 0.01939 eV/atom (95% interval [-0.02417,-0.01455] for the signed difference) and force RMS by 0.3926 eV/Å ([−0.4696,−0.3263]). Both runs improve. Graph validity remains similar: force minus control is +0.39 percentage points, with interval [−1.33,2.19]. Because the original force-dependent widths and candidate-selection rule remain fixed, this ablation tests the displacement itself, not the removal of all force information.

Q.2 MATCHED EXPERIMENT ON SOURCES AND PHYSICAL ADAPTATION

To complete the four-setting comparison, we construct physical targets from the Gaussian model’s own outputs using the same procedure as for harmonic. In each run, the frozen Gaussian model generates 32 anchors for each of the same eight training compositions. We record all 256 attempts and discard invalid anchors without replacing them. We retain the eight-candidate force-escort procedure, temperature, perturbation bounds, source seeds, and 1,000-update training schedule. For each physical student, we train a matched replay student from the same Gaussian base and add their parameter difference to that base with coefficient one. Within each physical/replay pair, reference examples, source randomness, training times, and teacher-selection draws are paired.

The two Gaussian runs retain 160 and 138 anchors after candidate-support checks. Teacher preparation uses 5,276 raw eSEN queries, including complete-work diagnostics stored by the original procedure. The students themselves sample uniformly from the allowed force-shifted candidates. We preserve the original maximum preparation budget, but actual valid-anchor counts and query counts can differ by source. Four new student fits of 1,000 updates produce the two paired Gaussian models. The harmonic physical model and the two unadapted models reuse their previously audited checkpoints and outputs.

The Gaussian physical models generate 1,536 unoptimized outputs on the existing 24-composition set at 128 network calls per attempt. eSEN evaluation adds 3,072 queries, and GFN2 adds 1,632 attempts including reference inversion checks. One GFN2 calculation fails on a graph-invalid output; every graph-valid output and reference calculation succeeds. The experiment’s audit reconstructs target proposals and graph restrictions, replays uniform candidate draws and parameter arithmetic, and reparses the structural and physical scores. This is a fixed follow-up on a previously evaluated set, not a new held-out composition test.

For Gaussian, physical adaptation changes energy on paired, jointly valid outputs by −0.01596 eV/atom under eSEN (95% interval [−0.02182,−0.01021]) and −0.01465 under GFN2 ([−0.02041,−0.00929]). The corresponding force-RMS changes are −0.3795 eV/Å ([−0.4618,−0.3045]) and −0.4618 ([−0.6195,−0.3370]). The paired physical update therefore also benefits the Gaussian-source generator.

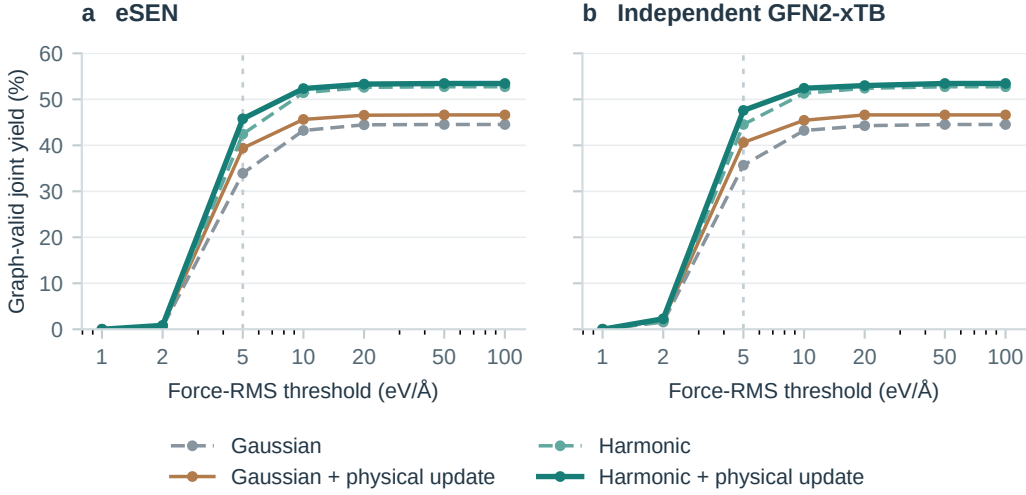


Figure 10: Joint graph-validity and residual-force yield for both sources, with and without the paired physical update. The denominator includes every raw generation attempt. Each threshold is applied to the RMS atomic-force magnitude in $\text{eV}/\text{\AA}$, and both potentials use the same threshold grid. No coordinates are optimized before evaluation.

At a force-RMS threshold of $5 \text{ eV}/\text{\AA}$, GFN2 joint-yield improvements in runs 1 and 2 are 4.56 and 5.34 percentage points for Gaussian and 3.13 and 2.99 for harmonic. After physical adaptation, harmonic minus Gaussian is 16.28 points in run 1 and -2.34 points in run 2, despite a positive pooled difference. Thus the results show both separate source/update benefits and variability between runs. They do not establish a positive statistical interaction between the source and physical update.

R CURVATURE-AWARE LOCAL WORK SAMPLING

The force escort translates every local candidate by the same vector. It does not change the shape of the proposal cloud. Molecular energy can vary more sharply in some directions, such as bond stretching, than in softer collective motions. We test whether estimating this local curvature produces better importance weights at a fixed candidate-query budget, without changing the target distribution.

Fitting local curvature. Flatten the anchor into $3N$ coordinates. Let the columns of $Q_N \in \mathbb{R}^{3N \times d}$ form an orthonormal basis for the centered space, with $d = 3(N - 1)$. A displacement $u \in \mathbb{R}^d$ is expressed in angstroms in this basis, and $Q_N u$ gives the corresponding atom-coordinate displacement. Let f_A be the anchor force in the same basis.

We draw eight independent pilot candidates. Their displacements form columns of D , and the negative changes in force form columns of G . We approximate $G \approx B_A D$ by fitting a symmetric matrix:

$$\|BD - G\|_F^2 + \lambda \|B - \kappa I\|_F^2, \quad \kappa = \max\left(0, \frac{\langle D, G \rangle_F}{\|D\|_F^2}\right), \quad \lambda = 0.01 \|D\|_F^2 / K. \quad (20)$$

Here K counts pilot columns that satisfy the graph restriction. If no pilot survives, we set $B_A = 0$ and recover the original force translation. For a nonempty pilot we floor $\|D\|_F^2$ at 10^{-20} in the scalar and ridge calculations; exactly zero displacements also give zero fitted curvature. The scalar κ estimates an average nonnegative stiffness. The second term shrinks poorly determined directions toward this isotropic estimate. We replace negative eigenvalues of the fitted matrix by zero to obtain a positive semidefinite B_A . The isotropic control uses $B_A = \kappa I$ directly. Pilot information and the fitted map are fixed before drawing production candidates.

Changing the proposal mean and covariance. The fitted quadratic energy is $E_+(A + Q_N u) - E_+(A) \approx -f_A^\top u + u^\top B_A u / 2$. Multiplying its Boltzmann factor by the original Gaussian ref-

erence gives another Gaussian with a shifted mean and adjusted covariance. Define $P_A = I + \sigma_A^2 B_A / (k_B T_{\text{phys}})$, a dimensionless positive-definite precision matrix. We map the source displacement u to

$$v = P_A^{-1/2} u + b_A, \quad b_A = P_A^{-1} \frac{\sigma_A^2 f_A}{k_B T_{\text{phys}}}, \quad \log J_A = -\frac{1}{2} \log \det P_A. \quad (21)$$

The factor $P_A^{-1/2}$ contracts perturbations more in stiff directions, while b_A adjusts the force-guided mean shift for curvature. The Jacobian J_A measures volume change in the d independent coordinates. We keep the original reference width and temperature, so only the proposal changes, not the restrained target. With $X = A + Q_N u$ and $Y = A + Q_N v$, complete work is

$$W_A = E_+(Y) - E_+(A) + k_B T_{\text{phys}} [\log q_A(X) - \log q_A(Y) - \log J_A]. \quad (22)$$

Candidates outside the original same-graph support receive zero weight. This is a local use of targeted and escorted work reweighting (Jarzynski, 2002; Vaikuntanathan & Jarzynski, 2008).

Why curvature can improve the weights. Write the actual energy as the fitted quadratic plus a remainder $R_A(v)$. Substitution gives $W_A = -k_B T_{\text{phys}} \log Z_{\text{quad}} + R_A(v)$, where Z_{quad} is the integral of the Gaussian reference times the quadratic Boltzmann factor. The fitted quadratic cancels from the varying part of the work; only the approximation error changes across allowed candidates. If energy is exactly that quadratic, all allowed candidates have equal weights. Tests verify this constant-work identity for an anisotropic quadratic energy and check rotation equivariance of the fitted map.

Pilot comparison at equal query budgets. For each of eight existing training compositions in each run, we take the first four eligible anchors without ranking by energy or ESS. The translation baseline uses 24 fresh production candidates per anchor. Each curvature-fitted map uses at most eight pilot energy/force queries and 16 fresh production candidates. The maximum particle-query budget is therefore the same. Common anchor queries and reflection evaluations are counted separately. We evaluate all production energies, including those of graph-invalid candidates, and do not reuse pilot samples as production samples for the fitted maps.

Table 26: Local ESS with complete-work weights at a maximum of 24 particle queries per anchor. Each run uses 32 anchors. Curvature methods reserve eight queries for fitting and use 16 production candidates; translation uses all 24 for production.

Escort	Production particles	ESS, run 1	ESS, run 2
Force translation	24	2.22	1.86
Isotropic curvature	16	3.55	3.94
Directional curvature	16	6.69	6.70

Directional curvature improves ESS over translation on 61 of 64 anchors. The mean increase is 4.66 effective samples, with composition-bootstrap 95% interval [3.81, 5.46]. Same-graph retention is 99.51%, compared with 99.54% for translation. Independent multivariate-Gaussian density calculations reproduce the stored complete-work weights. We then construct full teacher pools for all 349 original eligible anchors using fresh production draws. Translation versus curvature gives mean ESS of 2.32 versus 6.71 in run 1 and 2.33 versus 6.37 in run 2. These results concern the local teacher; the next experiment tests whether they improve the trained generator.

Does the teacher improvement transfer to generation? We train three physical students: uniform sampling of 24 force-translated candidates, complete-work weighting of those same 24 candidates, and complete-work weighting of 16 curvature-adjusted candidates. Each uses the original 1,000-update schedule and reference/anchor choices. We reuse the matched replay model and add the physical-minus-replay parameter difference with coefficient one. All 349 original anchors remain eligible. The two work-weighted methods have equal maximum particle-query budgets; uniform force targets can omit candidate-energy queries. This follow-up changes the candidate-pool size and curvature fit. The main experiments continue to use the original, preselected eight-candidate force model.

Table 27: Generated outputs after training with the three teachers, evaluated on 24 compositions with 1,536 attempts per method across both runs. Energy changes compare a work-weighted student with the uniform-force student on paired, jointly valid outputs. All methods use 128 network evaluations and no geometry optimization.

Teacher	Graph validity (%)	eSEN change (eV/atom)	GFN2 change (eV/atom)
Uniform force, 24 candidates	54.30	0	0
Complete work, 24 candidates	53.39	+0.00959	+0.00845
Curvature work, 16 candidates	53.32	+0.00793	+0.00783

Curvature work minus translation work changes energy by -0.00184 eV/atom under eSEN (95% interval $[-0.00654, 0.00289]$) and -0.00068 under GFN2 ($[-0.00484, 0.00341]$). The two runs have opposite signs. The higher teacher ESS therefore does not establish an additional generation benefit. Uniform force targets give lower energy and force RMS than either work-weighted student under both potentials. Relative to the frozen harmonic generator, this 24-candidate uniform-force model lowers energy by 0.02088 eV/atom under eSEN and 0.01918 under GFN2. The primary method retains the inexpensive force-only recipe.